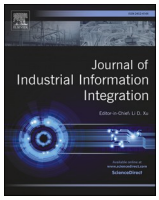
Journal of Industrial Information Integration 50 (2026) 101067



Contents lists available at ScienceDirect

Journal of Industrial Information Integration

journal homepage: www.sciencedirect.com/journal/journal-of-industrial-information-integration



Full Length Article



Multi-dimensional framework for assessing digital twin maturity in construction machinery

Ruibo Hu, Wenting Gong, Ke Chen *, Hanbin Luo

School of Civil and Hydraulic Engineering, Huazhong University of Science and Technology, Wuhan, China

ARTICLE INFO

Keywords:
Construction machinery
Digital twin
Maturity model
Maturity assessment

ABSTRACT

Smart construction machinery enabled by digital twin (DT) technology has significant potential to enhance construction safety and efficiency. However, the lack of a dedicated maturity assessment model for construction machinery DT (CMDT) reveals a critical gap in existing research. This study proposes a structured framework for assessing CMDT maturity, designed to evaluate the readiness and developmental stage of DT implementations in construction machinery. The framework was developed based on a comprehensive literature review and expert interviews, yielding a maturity assessment model comprising five dimensions and 24 indicators across five maturity levels. The model adopts a two-stage assessment approach that integrates expert competency-based grouping and weighting with an improved bi-objective optimization model. To enhance the robustness of expert opinion aggregation, a penalty-weight mechanism is embedded in the objective function, effectively balancing consensus and confidence. The proposed framework was validated through a real-world case study involving an automated construction system (ACS). The results demonstrate the capability of the framework to identify CMDT maturity levels and inform improvement pathways. Overall, this study provides an evidence-based tool to accelerate DT adoption in the construction sector.

1. Introduction

As the construction industry advances toward mechanization and digitalization, sophisticated machinery such as tunnel boring machines (TBMs) for shield tunnels [1] and automated construction systems (ACSs) for high-rise buildings [2] has been increasingly deployed across a wide range of projects. These machines offer substantial advantages, but their structural complexity, harsh on-site environments, and significant fluctuations in load conditions pose considerable challenges, which necessitate the adoption of digital technologies to enhance operational safety and efficiency [3]. Among the available digital technologies, digital twin (DT) technology has proven instrumental in facilitating the operation of construction machinery [4]. DTs can integrate geometric and non-geometric data at both component and system levels, enabling closed-loop processes for monitoring, prediction, and optimization [5–6]. These capabilities have generated growing interest in DT applications within the construction sector, as evidenced by several review studies [7–9].

As DT technology continues to mature, evaluating the maturity of DT implementation has become essential for assessing the current state of applications and for devising targeted improvement strategies [10]. Several maturity models have been proposed to address this need. For instance, Uhlenkamp et al. [11] developed a general maturity model encompassing multiple dimensions, including model, data, context, control, human-machine interaction, computing capabilities, and integration. Liu et al. [12] proposed general maturity models applicable across multiple domains. However, such generalized models often lack the technical granularity required to capture domain-specific characteristics.

Recognizing the distinction between general and domain-specific DT applications, Hu et al. [13] developed a quantitative maturity model for high-end equipment, featuring 27 rubrics across three primary dimensions. Chen et al. [14] and Li et al. [15] designed maturity assessment frameworks for building and infrastructure DTs, respectively. Despite these advancements, existing DT maturity models do not sufficiently account for the unstructured nature of construction environments or the inherent complexity of large construction machinery. The absence of tailored maturity models has left the improvement pathways of these applications largely unclear.

To address these limitations, this study proposes a domain-specific

* Corresponding author.
E-mail address: chenkecm@hust.edu.cn (K. Chen).

https://doi.org/10.1016/j.jii.2026.101067

Available online 13 January 2026
2452-414X/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

maturity assessment framework for construction machinery digital twins (CMDT). It pursues two primary objectives. First, it develops a multi-dimensional maturity model incorporating a customized indicator system and evolutionary pathways. Second, it introduces a novel assessment methodology that accounts for indicator importance and expert competency. The main contributions of this study are summarized as follows:

- A domain-specific maturity model for CMDT is proposed. The model comprises five dimensions and 24 indicators, reflecting the unique technical and operational complexities of construction machinery.
- A robust maturity assessment methodology is developed by integrating Louvain-based community detection for expert grouping with an improved bi-objective optimization model for opinion aggregation. This approach simultaneously accounts for differences in expert competency and the quality of consensus, thereby enhancing the reliability of the assessment results.
- The assessment results provide evidence-based decision support for developing strategic implementation roadmaps for CMDT.

The remainder of this paper is organized as follows. Section 2 reviews the existing approaches to DT maturity measurement. Section 3 presents the proposed CMDT maturity assessment framework. Section 4 presents a case study that highlights the critical issues in DT maturity measurement. Section 5 discusses the results, and Section 6 concludes the paper and outlines directions for future research.

2. Literature review

To structure the existing literature on DT maturity, a targeted literature search was conducted. The Scopus and Google Scholar databases were queried using the combined keywords "Digital Twin" AND "Maturity", covering publications from 2010 to 2025. The review is organized into two subsections. Section 2.1 synthesizes existing DT maturity models, focusing on their core attributes, evaluation dimensions, and evolutionary pathways. Section 2.2 summarizes qualitative and quantitative approaches to DT maturity assessment.

2.1. DT maturity models

Technology maturity denotes the stage of development that a technology has reached within its life cycle [16]. As the scope of DT applications continues to expand, research on maturity models has attracted growing attention. A general DT maturity model provides a structured framework for identifying the attributes of DT systems and defining corresponding evaluation criteria. The key attributes of DTs include integration, connectivity, analytics, security, and scalability. Integration refers to the ability to aggregate data from multiple sources and is commonly evaluated using metrics such as data completeness, consistency, and integration latency [17]. Connectivity denotes the capability to seamlessly link physical and digital environments, which can be assessed through connectivity uptime, interoperability scores, protocol compatibility, and integration complexity [18]. Analytics reflects proficiency in data analysis and decision support, with indicators including decision-making speed and effectiveness [19]. Security concerns the protection of data privacy, which can be evaluated using metrics such as the frequency of security incidents, adherence to security standards, and the robustness of encryption protocols [20]. Finally, scalability assesses the ability to handle increasing data volumes and complexities while maintaining stable performance [21].

Building on these foundational attributes, several DT maturity models have been proposed. For example, Liu et al. [12] introduced a five-level maturity model, progressing from digital models to federated DTs. Similarly, a six-level DT maturity model has been introduced, advancing from asset replication to interoperability [22]. Beyond general models, domain-specific DT maturity models cater to the unique requirements of different sectors. In the aerospace industry, Medina et al. [23] proposed a 10-dimensional DT maturity model with four maturity levels per dimension, covering aspects such as the analytical capability and data collection frequency. In the manufacturing industry, Villegas et al. [24] introduced a DT maturity model, focusing on product, production, and service dimensions. For human-robot collaboration in assembly tasks, Liu et al. [25] designed a maturity model featuring five dimensions and 24 indicators. Domain-specific models have also been developed for healthcare [26], smart cities [27], and underground infrastructure [15], each presenting tailored dimensions, indicators, and maturity levels. However, considering the inherent complexity of large construction machinery, a tailored DT maturity model is urgently required. In the absence of such a model, project managers will face challenges in formulating effective DT implementation strategies, identifying improvement priorities, and leveraging competitive advantages.

2.2. Maturity assessment methods

Ensuring the reliability of DT maturity assessments necessitates the selection of appropriate methods. Generally, maturity assessment methods can be divided into qualitative and quantitative categories.

In the qualitative category, online questionnaires [28], face-to-face interviews [29], and site investigations [30] are commonly used to collect expert opinions on maturity indicators. The selection and grouping of decision-makers (DMs) play a critical role in this process. Previous studies have engaged varying numbers of experts, such as 9 in [25], 15 in [31], and 22 in [15]. According to group decision-making theory, when the number of DMs exceeds 11, their aggregated opinions yield more accurate and reliable evaluations [14]. Additionally, DMs are often grouped based on their homogeneity in knowledge and experience with DT implementations. Higher weights are assigned to groups with greater expertise [32]. For example, Chen et al. [33] applied the K-means clustering algorithm to group 20 DMs by expertise level and employed the Technique for Order of Preference by Similarity to Ideal Solution to assign higher weights to expert groups. Other studies have also employed K-means clustering for expert grouping [34]. However, traditional K-means clustering relies on manually specified group numbers, which may introduce bias [35]. Moreover, in addition to expertise levels, factors such as attentiveness and the quality of responses warrant consideration when assigning weights [36]. For subsequent quantitative analysis, maturity levels in questionnaires are typically measured using data points (e.g., 0 or 1) [27] or intervals (e.g., [0, 1] or [1, 2]) [33]. Interval-based measurements mitigate ambiguities in human judgment, yielding more objective evaluations [37].

In the quantitative category, individual opinions are first quantified. Traditionally, these opinions are represented numerically, as demonstrated in [38]. However, because human judgment inherently involves imprecision, probability distribution functions (PDFs) are increasingly employed. For instance, by observing the characteristics of data distribution, each opinion regarding the maturity level of an indicator can be fitted to a Gaussian [33] or Beta distribution [39]. The average value of the PDF represents the expert′s opinion, and the standard error quantifies the uncertainty associated with the judgment. Once individual opinions are quantified, aggregation methods maximize the total overlapping area of PDFs, converting it into an optimization problem, as shown in [40] and [41]. To further enhance the reliability and objectivity of the results, Ji et al. [39] proposed a bi-objective optimization model that considers both the overlapping area of the PDFs and the confidence level of the collective PDF. Subsequent adaptations, such as those applied to building information modeling (BIM) maturity assessment [31] and human-robot collaboration maturity assessment [25], illustrate the model′s utility. However, in the original bi-objective optimization model, maximizing the overlapping area of PDFs and minimizing the confidence level of the collective PDF are weighted equally, which can lead to biases that undervalue consensus accuracy.

As summarized in Table 1, although existing maturity models have advanced the evaluation of DTs, two primary research gaps remain. First, most existing models are either cross-domain or primarily tailored to manufacturing applications, and thus fail to adequately capture the unstructured, dynamic, and harsh operating characteristics of construction machinery. Second, many maturity assessment approaches do not explicitly consider indicator importance or differences in expert competency, which may reduce the reliability of evaluation outcomes. To address these limitations, this study proposes a domain-specific maturity assessment framework for CMDT. The framework combines a customized 24-indicator evaluation system with a bi-objective optimization model that incorporates PDFs, thereby yielding more objective and reliable maturity assessment results.

3. Framework for assessing the maturity of CMDT

The proposed CMDT maturity assessment framework contains two phases, as illustrated in Fig. 1. Phase 1 focuses on maturity model development, in which domain-specific indicators were identified through systematic indicator selection and subsequently prioritized through importance ranking. This phase is completed by defining discrete maturity levels for evaluation. Phase 2 establishes a rigorous maturity evaluation methodology that integrates Louvain-based community detection for expert grouping with an improved bi-objective optimization model for opinion aggregation. By balancing consensus accuracy and confidence levels through a penalty-weight mechanism, the inherent uncertainty associated with expert assessments can be mitigated. Consequently, the proposed framework can identify current CMDT maturity levels and support strategic DT implementation.

3.1. CMDT maturity model

Recognizing the unique characteristics of construction machinery, the CMDT maturity model can be expressed as:

$$CM_{DT} = (PE,\ VM,\ DD,\ CI,\ SE) \tag{1}$$

where

PE (Physical Entity) represents the construction machinery itself, focusing on its integration with sensors, actuators, and real-time data collection systems.

VM (Virtual Model) refers to the virtual representation of the machinery, emphasizing the accuracy of replication and performance simulation capabilities.

DD (DT Data) covers data-related aspects, including storage, integration, and utilization.

CI (Connection and Interaction) addresses the integration between physical machinery, virtual twins, and the broader data ecosystem, including human–machine interaction enabled by technologies such as augmented and virtual reality.

SE (Service) encompasses services such as predictive maintenance, parameter optimization, operational decision-making, and health monitoring.

The indicators for each dimension were derived from peer-reviewed journal papers, and the KH Coder software package was used to reduce subjectivity. The collected indicators are detailed in Table 2.

To minimize subjectivity in indicator selection, a structured questionary survey was conducted. Survey participants were required to meet at least one of the following qualifications: 1) possession of professional credentials in construction machinery; 2) direct involvement in DT-related projects, including roles in system design, implementation, or operational management; or 3) a proven research track record, as evidenced by academic publications pertaining to DT. The first part of the questionary assessed the competency of the experts to ensure that the opinions of more qualified experts were given greater weight. Expert competency (EA_j) was calculated using the following formula:

$$EA_j = \frac{\sum_Q S_q}{Q} \tag{2}$$

where S_q represents the score for the q-th question on the expert's competency and Q denotes the total number of questions assessing expertise. This approach ensured that evaluations from highly competent experts significantly influenced the determination of indicator importance.

The second part of the questionary focused on the relative importance of individual indicators. Experts rated each indicator using a 5-point Likert scale (1 = Not important, 5 = Very important). A weighted scoring method was used to calculate the importance score of each indicator (v_i):

$$v_i = \frac{\sum_{j=1}^{n} I_{ij} \times EA_j}{\sum_{j=1}^{n} EA_j} \tag{3}$$

where I_{ij} is the score assigned to the i th indicator by the j-th expert and n is the total number of experts.

The questionary survey was delivered via the SoJump platform (https://www.wjx.cn/vm/enzjcB1.aspx). Responses from 20 experts were obtained for analysis. Using the weighted scoring method, the importance of each indicator was calculated and ranked, as illustrated in Fig. 2. The bottom four indicators, including user experience of interaction, utilization of immersive technologies, service scope, and fleet management, have the lowest importance. Subsequent discussions with experts provided the following insights: 1) the utilization of immersive technologies is not widely applied in current DT implementations; 2) user experience of interaction is closely related to human–machine interaction, rendering it redundant; 3) service scope overlaps with other service-related indicators in the SE dimension; and 4) fleet management is not a primary focus for construction machinery. Consequently, these four indicators were excluded from the final indicator system. The remaining indicators were categorized into quartiles based on their importance scores. The indicators in the first, second, third, and fourth

Table 1
Summary of representative works and comparison with the proposed study.

Studies	Domains	DT maturity models			Maturity assessment methods			
		Dimensions	Indicators	Maturity levels		Indicator weights	Expert weights	Objective weights
Uhlenkamp et al. [11]	Cross-domain	7	31	×	P airwise comparison	×	×	×
Liu et al. [12]	Cross-domain	7	×	5	Qualitative analysis	×	×	×
Hu et al. [13]	High-end equipment	3	27	6	Analytic hierarchy process	√	×	×
Medina et al. [23]	Aerospace	10	×	4	Interviews and scoring	×	×	×
Cimini et al. [56]	Manufacturing	3	16	4	Maximum score method	×	×	×
Chen et al. [33]	BIM	3	27	10	Quantitative measurement model	√	√	×
This study	Construction machinery	5	24	5	Opinion aggregation based on PDFs	√	√	√

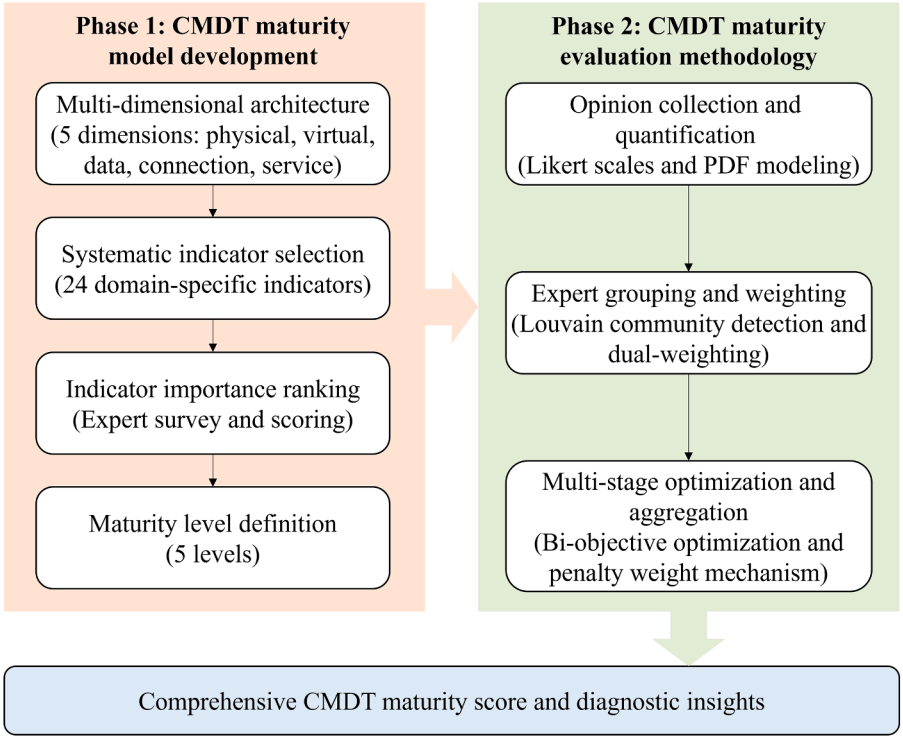


Fig. 1. Overview of the proposed framework.

quartiles were designated as critical importance (L1), high importance (L2), moderate importance (L3), and low importance (L4), respectively.

After determining the maturity indicators, a five-level maturity structure was developed based on established DT maturity models [13–15,22,47,50,51]. The five maturity levels are defined as follows:

Level 1. Basic monitoring

At this foundational level, construction machinery exhibits minimal digital integration. Sensors, if installed, are rudimentary, and data are collected manually in spreadsheets or static formats. Real-time virtual representations of the machinery are absent, with any 3D models being static and used solely for visualization. Machinery operates independently, without remote-control capabilities or digital services, relying entirely on manual oversight and intervention.

Level 2. Connected machinery

At this level, construction machinery incorporates basic sensors to track fundamental metrics, such as fuel consumption and engine temperature. A rudimentary DT reflects the real-time status of the machinery for a limited set of parameters. Structured data integration supports basic monitoring and control capabilities. Digital services are introduced, including notifications for low-fuel or maintenance alerts, allowing for minimal remote intervention by operators.

Level 3. Predictive machinery

At this level, construction machinery is equipped with advanced sensors and actuators that capture real-time data on key operational parameters, such as load, pressure, and component wear. This level signifies the achievement of perception intelligence, wherein machines are capable of task-specific recognition and basic contextual awareness. Centralized data integration supports the application of conventional machine learning algorithms for predictive analytics. Key functionalities at this level include predictive maintenance and elementary fleet management.

Level 4. Semi-autonomous machinery

At this level, construction machinery demonstrates cognitive intelligence by dynamically adjusting operational parameters in response to real-time data and environmental conditions. This cognitive capability is supported by continuous simulations performed by the DT model, which can optimize performance in real time by utilizing more advanced machine learning methods. Advanced data management and predictive analytics enable services such as automated fleet management, optimized scheduling, and remote operation.

Level 5. Fully autonomous machinery

At this highest level, construction machinery achieves autonomous intelligence with minimal or no human intervention. The DT model matures into a fully self-adaptive system, leveraging advanced AI technologies, including large language models and multi-agent coordination frameworks. These technologies enable machinery to continuously learn from both historical data and real-time feedback. Machinery autonomously adjusts parameters to enhance efficiency and mitigate risks. Additionally, data management becomes fully automated, supporting seamless integration of processes. This level introduces advanced digital services such as autonomous fleet management and lifecycle optimization.

After establishing the five levels of CMDT maturity, the maturity levels for each individual indicator were further specified. Detailed definitions for the five-level maturity scale at the indicator level are provided in Appendix A to ensure consistency and reproducibility of the assessment.

3.2. CMDT maturity assessment

Fig. 3 shows the process of CMDT maturity assessment, which integrates expert judgment with quantitative analysis. The process begins with expert evaluations, in which each expert's assessment of the indicators is represented as an individual probability distribution. Experts are then grouped according to their competency, with those holding similar opinions clustered together. Within each group, a composite probability distribution is generated to represent the collective

Table 2
Identified indicators.

Dimension	Indicators	References	Explanation
PE	Sensor diversity	[15,25,27, 42-44]	Variety of sensors monitoring different aspects of the machinery.
	Sensor coverage	[15,42,25, 27]	Distribution and quantity of sensors ensuring comprehensive data collection.
	Data acquisition	[13,23,25, 44-46]	Real-time data acquisition (e.g., load, speed, and hydraulic pressure).
	Adaptive actuators	[22,43]	Self-adjusting actuators preventing overload or inefficiencies.
	Level of autonomy	[22,44-48]	Autonomous operation without human intervention.
VM	Model fidelity	[13-15,25, 42,47]	High-fidelity virtual representation of physical machinery.
	Model update	[14,15,23, 25,27,47]	Synchronization mechanism and frequency between the virtual model and physical machinery.
	Simulation	[23,43,44, 48]	Capability of the virtual model to simulate scenarios based on current operational data.
	Predictive analytics	[15,23,43, 47,48]	Predictive capability for forecasting issues such as wear and tear based on historical and real-time data.
	Continuous learning	[13,25,27, 43,47]	Learning from historical data to enhance performance.
	Model interoperability	[14,15,23, 25,42,47]	Capability to integrate with other systems and models to support multi-scale, multi-physical, and multi-level model expansion.
DD	Data completeness	[13,15,25, 27,42,44,47]	All required data points are collected and stored without omissions.
	Data quality	[13,14,25, 42,45,47]	Accuracy, consistency, and reliability of the data stored.
	Data standardization	[13-15,25, 27,42]	Consistent data formats and definitions for enhanced interoperability.
	Real-time data processing	[42,45,48]	Capability to process and update data in real time for actionable insights.
	Data accessibility	[23,25,27, 42]	The extent to which users can access and retrieve the required data.
	Data security	[27,45]	Measures and protocols in place to protect data from unauthorized access, breaches, and loss.
CI	System connectivity	[14,22,23, 27,47]	The extent to which physical machinery and the DT are connected.
	Real-time feedback loop	[13,14,22, 23,25,42,44, 46,48]	Capability of the virtual model to send real-time feedback to the physical machinery for adjustments.
	Cloud integration	[14,25,27, 43]	Capability to allow remote monitoring, control, and data sharing between the physical and virtual worlds.
	Human–machine interaction	[13,23,25, 27,47]	Level of human interaction with machinery for operation, monitoring, and training.
	Utilization of immersive technologies	[13,27,43, 46]	Capability to enhance the interaction and visualization of data from the physical machinery and its virtual counterpart.
	User experience of interactions	[47,49]	Satisfactory level to users interacting with the DT.
SE	Service scope	[14,25,42]	Range of services provided, from basic monitoring to advanced autonomous services.
	Predictive maintenance	[43,46]	Capability to predict and schedule maintenance based on real-time data, reducing downtime and improving machinery longevity.
	Operational optimization	[13,15,23, 43-46]	Capability to optimize machinery operations based on environmental factors.
	Safety and risk management	[44,48]	Capability to identify, monitor, predict, and mitigate safety risks and hazards to protect personnel, machinery, and the environment.
	Fleet management	[15,46]	Process of optimizing the operation, maintenance, and utilization of a fleet of construction machinery.

judgment while accounting for intra-group variability. Subsequently, the group-specific distributions are aggregated into an overall probability distribution by incorporating the relative weights of each cluster, yielding a comprehensive evaluation of each indicator. Finally, the overall CMDT maturity level is determined through analysis of the aggregated distributions.

3.2.1. Opinion collection and quantification

The questionnaire for expert evaluation comprises two sections: the first captures respondents′ expertise levels, and the second captures their assessments of specific indicators. The expertise levels determine the significance of each respondent′s input, with a greater weight assigned to participants possessing higher expertise. Indicator evaluations are scored on a five-point Likert scale (1–5) and mapped to five intervals: [0, 1), [1, 2), [2, 3), [3, 4), and [4, 5]. This interval-based approach effectively captures uncertainty and imprecision in human judgments, providing nuanced insights compared to single-value scores [14].

Given n participants, the responses set is denoted as $\boldsymbol{R} = \{R_1, R_2, \ldots, R_i, \ldots, R_n\}$, where R_i represents the response of the i th respondent. Each respondent completes $(p + m)$ questions, with $R_i = \{E_i, L_i\}$. $E_i = \{e_{i1}, e_{i2}, \ldots, e_{it}, \ldots, e_{ip}\}$, indicating the i th respondent′s answers to p questions on expertise, where $t = 1, 2, \ldots, p$. $L_i = \{l_{i1}, l_{i2}, \ldots, l_{ih}, \ldots, l_{im}\}$, indicating the i th respondent′s answers to m questions on indicator levels, where $h = 1, 2, \ldots, m$.

Expert evaluations often exhibit subjective or ambiguous elements, with their confidence levels varying based on individual judgment. To quantify this uncertainty, probability theory is applied. Each respondent′s opinion for a given indicator is modeled as a PDF. Probabilities are assigned to the five intervals of indicator levels, with responses typically following a generalized Gaussian distribution [25]:

$$f_i^h(x;\mu,\sigma,y_0) = y_0 + \frac{1}{\sigma\sqrt{2\pi}} * e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{4}$$

$$F_i^h(x) = \int f_i^h(x)dx = 1 \tag{5}$$

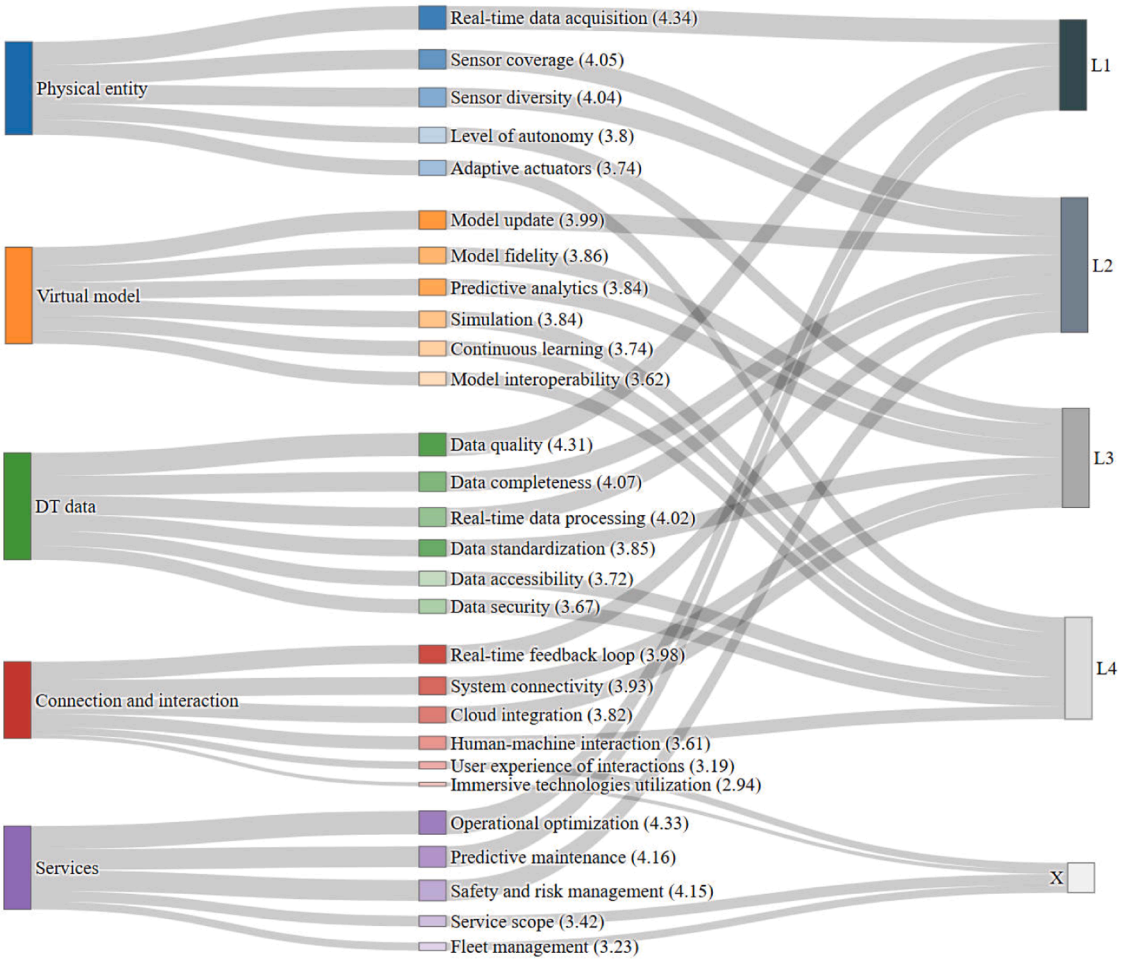


Fig. 2. Sankey diagram of indicators. X denotes indicators excluded from the final indicator system.

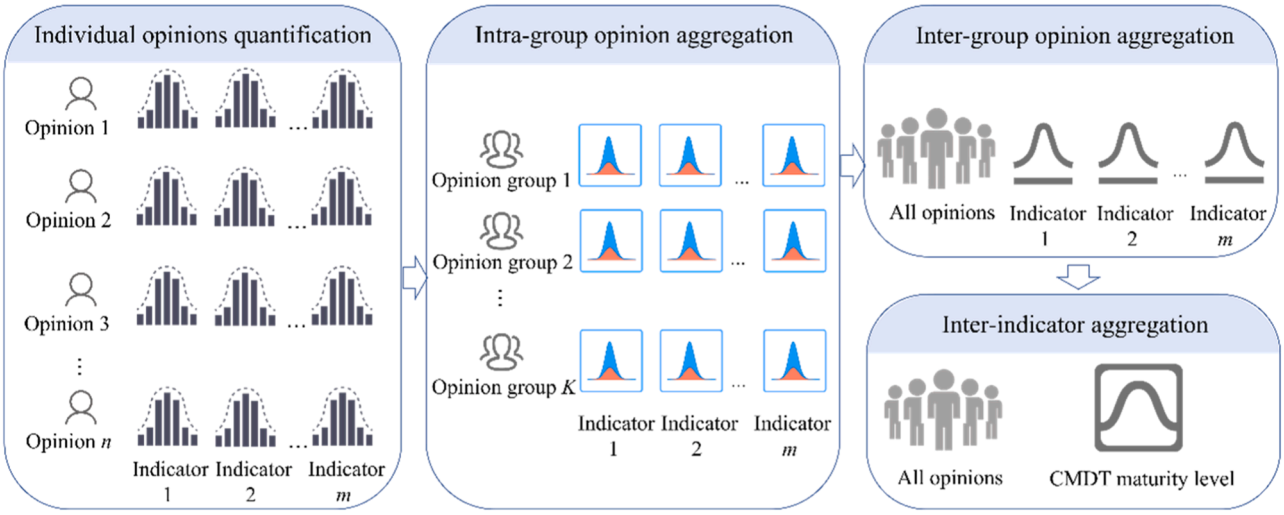


Fig. 3. Four-stage opinion aggregation process for CMDT maturity assessment.

where $f_i^h(x)$ is the PDF for the h-th indicator of the i th respondent, x represents the indicator level, μ is the mean value, and σ is the standard deviation, indicating the degree of uncertainty. $F_i^h(x)$ is the cumulative distribution function of $f_i^h(x)$. This probabilistic approach provides a flexible and robust quantification of respondents' opinions, incorporating both central tendency and confidence levels.

3.2.2. *Expert opinion grouping*

Instead of seeking consensus across all individuals, achieving consensus within more homogeneous groups enhances the reliability of analysis [52]. Respondents are grouped based on expertise metrics using

a community detection approach that addresses the overlapping characteristics of expertise profiles. Compared to traditional clustering methods, community detection dynamically identifies the optimal number of communities and effectively handles overlapping areas caused by the interrelated nature of expertise metrics. This study employs Louvain-based community detection for grouping opinions. First, expertise metrics are standardized for comparability across respondents:

$$\widehat{e}_{it} = \frac{e_{it} - \min(e_t)}{\max(e_t) - \min(e_t)} \tag{6}$$

where $\min(e_t)$ and $\max(e_t)$ are the minimum and maximum values of metric t, respectively.

Second, cosine similarity S_{ij} is calculated for each pair of opinions (i, j) based on their normalized expertise metrics:

$$S_{ij} = \frac{\sum_{t=1}^{p} \widehat{e}_{it} \cdot \widehat{e}_{jt}}{\sqrt{\sum_{t=1}^{p} \widehat{e}_{it}^2} \cdot \sqrt{\sum_{t=1}^{p} \widehat{e}_{jt}^2}} \tag{7}$$

where S_{ij} ranges from 0 (no similarity) to 1 (identical expertise profiles).

Third, each respondent i is represented as a node in the network. An edge is added between nodes i and j with a weight S_{ij}, provided their similarity exceeds a threshold θ (optional, to reduce weak connections). A weighted adjacency matrix A is constructed such that $A_{ij} = S_{ij}$ for connected pairs and $A_{ij} = 0$ for unconnected nodes. Finally, communities are identified by maximizing modularity Q:

$$Q = \frac{1}{2d} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2d} \right) \delta(c_i, c_i) \tag{8}$$

where d is the sum of all edge weights, $k_i = \sum_j A_{ij}$ is the weighted degree of node i, $\delta(c_i, c_i)$ equals 1 if nodes i and j belong to the same community and 0 otherwise.

The algorithm iteratively groups nodes to maximize Q, merging nodes into communities until no further improvement in modularity is achieved. The resulting communities represent clusters of respondents with similar expertise.

After grouping respondents, groups with higher levels of expertise should be assigned greater weights in the maturity assessment. However, even among groups with similar expertise levels, differences in consensus can arise owing to attention to detail and response quality. A lower consensus within a group reduces its reliability and influence on the final analysis [53]. To account for both expertise and consensus, a weighting strategy is proposed. For each group G_z, the average expertise score is computed as follows:

$$ES_{G_z} = \frac{1}{|G_z|} \sum_{i \in G_z} \sum_{t=1}^{p} e_{it} \tag{9}$$

where ES_{G_z} is the expertise score for group G_z, e_{it} denotes the t-th expertise metric of respondent i, and $|G_z|$ is the total number of respondents in G_z.

Fuzzy entropy $H(G_z)$ is calculated to quantify intra-group consensus. Lower entropy values indicate a higher consensus:

$$H(G_z) = -\sum_{h} \frac{C_h}{|G_z|} \log \left(\frac{C_h}{|G_z|} \right) \tag{10}$$

where C_h is the frequency of the most common responses for indicator h within G_z.

The final weight W_{G_z} for each group is determined by balancing expertise and consensus:

$$W_{G_z} = \alpha \cdot ES_{G_z} + \beta \cdot (1 - H(G_z)) \tag{11}$$

where α and β are coefficients that adjust the relative importance of expertise and consensus.

This weighting strategy ensures that groups with higher expertise and stronger consensus exert greater influence on maturity assessment, thereby enhancing the robustness of the evaluation process.

3.2.3. Opinion aggregation

The CMDT maturity assessment is derived through a four-stage opinion aggregation process: intra-group opinion aggregation, inter-group opinion aggregation, intra-dimension opinion aggregation, and inter-indicator opinion aggregation.

For the i th respondent's opinion, denoted as $f_i = \left\{ f_i^1, f_i^2, \ldots, f_i^m \right\}$, the intra-group PDF for group G_z is denoted as $f_{G_z} = \left\{ f_1, f_2, \ldots, f_i, f_{|G_z|} \right\}$. The aggregated intra-group PDF for the h-th indicator is modeled as:

$$f_{G_z}^h(x) = \varphi\left(f_i^h(x) \right), \ i = 1, 2, \ldots, |G_z| \tag{12}$$

where φ is the aggregated function used to combine individual opinions into a unified group opinion for the h-th indicator. The aggregation process considers the consensus and confidence levels. The consensus level is quantified as the overlapping area (O_A) of individual opinions, defined as:

$$O_A = \int \min \left\{ f_i^h(x), f_{G_z}^h(x) \right\} dx \tag{13}$$

The confidence level is formalized as the concentration of the PDF and is quantified by the variance σ^2, which reflects the degree of certainty or dispersion in the aggregated expert opinions. To more effectively balance the dual objectives of consensus accuracy and confidence level, a penalty weight λ is introduced to ensure that both aspects are jointly considered. In addition, the Quantile Averaging (QA) method is adopted to achieve optimal aggregation. This method ensures unimodal properties while incorporating the full range of expert opinions [54].

Accordingly, a bi-objective optimization problem is formulated. Formula (14a) maximizes a scalarized bi-objective. Normalized overlap $\overline{O_A}$ encourages agreement between the aggregated PDF and individual PDFs (consensus accuracy), while $\overline{\sigma^2}$ penalizes overly dispersed aggregated opinions (confidence level). Formula (14b) defines the QA aggregation in quantile space, which is subsequently transformed back to a PDF via differentiation. Formula (14c) defines O_A as the sum of overlap areas between each individual PDF and the QA-aggregated PDF. Formula (14d) specifies the mean and variance of the aggregated PDF. Formulas (14e) and (14f) apply min–max normalization to ensure comparability and numerical stability during optimization. Formula (14g) defines the decision variable constraints to ensure that all weights are non-negative and sum to unity. Formula (14h) defines the boundary constraints and operational domains of the variables and parameters involved in the optimization process.

$$\text{maxobj}\left(w_{G_z}^h \right) = \overline{O_A}\left(w_{G_z}^h \right) - \lambda \overline{\sigma^2}\left(w_{G_z}^h \right) \tag{14a}$$

$$\text{s.t.} f_{QA\ G_z}^h(x) = \frac{d\left(\left(\sum_{i=1}^{|G_z|} w_{iG_z}^h \left(F_i^h(x) \right)^{-1}(\rho) \right)^{-1} \right)}{dx} \tag{14b}$$

$$O_A\left(w_{G_z}^h \right) = \sum_{i=1}^{|G_z|} \int \min \left\{ f_i^h(x), f_{QA\ G_z}^h(x) \right\} dx \tag{14c}$$

$$\sigma^2\left(w_{G_z}^h \right) = \int \left(x - \int x f_{QA\ G_z}^h(x) dx \right)^2 f_{QA\ G_z}^h(x) dx \tag{14d}$$

$$\overline{O_A}\left(w_{G_z}^h \right) = \frac{O_A\left(w_{G_z}^h \right) - \text{Min}O_A\left(w_{G_z}^h \right)}{\text{Max}O_A\left(w_{G_z}^h \right) - \text{Min}O_A\left(w_{G_z}^h \right)} \tag{14e}$$

$$\overline{\sigma^2}\left(w^h_{G_z}\right) = \frac{\sigma^2\left(w^h_{G_z}\right) - \mathrm{Min}\sigma^2\left(w^h_{G_z}\right)}{\mathrm{Max}\sigma^2\left(w^h_{G_z}\right) - \mathrm{Min}\sigma^2\left(w^h_{G_z}\right)} \tag{14f}$$

$$\sum_{i=1}^{|G_z|} w^h_i = 1,\ w^h_i \geq 0,\ i = 1,\ 2,\ \ldots, |G_z| \tag{14g}$$

$$x \in [0,\ 5], \rho \in [0,1], \lambda \in [0,1] \tag{14h}$$

where the decision variable $w^h_{G_z} = \left\{w^h_{1G_z},\ w^h_{2G_z},\ \ldots,\ w^h_{iG_z}, w^h_{|G_z|G_z}\right\}$ is the weight vector for the h-th indicator in group G_z; x stands for the indicator level and the QA formula $f^h_{QA\ G_z}(x)$ is the aggregated opinion formula for group G_z; $\left(F^h_i(x)\right)^{-1}(\rho)$ is the inverse function of $F^h_i(x)$; and O_A and σ^2 are normalized as $(\overline{O_A})$ and $(\overline{\sigma^2})$ using min-max scaling for optimization, enabling conversion into a single-objective function.

The penalty parameter λ plays a critical role in balancing the objectives. If λ is significantly large, the optimization overly prioritizes minimizing σ^2, resulting in narrow fitted curves that fail to sufficiently overlap with the original PDFs. Conversely, if λ is significantly small, the optimization neglects σ^2, leading to excessively wide curves with reduced descriptive power. The Hippopotamus Optimization (HO) algorithm [55] was employed with key hyperparameters set as follows: the population size was set to 30 to preserve solution diversity, the maximum number of iterations was limited to 200 to provide adequate search depth, and the Lévy flight distribution index was set at 1.5 to facilitate global exploration. The optimal objective solution $w^{h*}_{G_z} = \left\{w^{h*}_{1G_z},\ w^{h*}_{2G_z},\ \ldots,\ w^{h*}_{iG_z}, w^{h*}_{|G_z|G_z}\right\}$ is obtained by solving Formula (14).

Upon incorporating all opinion groups, the optimal objective solution set $w^{h*} = \left(w^{h*}_1,\ w^{h*}_2,\ \ldots,\ w^{h*}_i,\ w^{h*}_n\right)$ is obtained. The optimized respondent weights $w^h = \left(w^h_1,\ w^h_2,\ \ldots,\ w^h_i,\ w^h_n\right)$ for each indicator can be calculated based on the group weights $W = (W_{G_1},\ W_{G_2},\ \ldots,\ W_{G_z},\ \ldots,\ W_{G_K})$, as follows:

$$w^h_i = w^{h*}_i \times\ W_{G_z},\ i \in G_z \tag{15}$$

The inter-group aggregation process combines the group-level opinions. The aggregated PDF for the h-th indicator across all K groups is modeled as

$$f^h(x) = \frac{d}{dx}\left[\left(\sum_{z=1}^{K} W_{G_z} \sum_{i=1}^{|G_z|} w^h_i\left[\left(F^h_i(x)\right)^{-1}(\rho)\right]\right)^{-1}\right] \tag{16}$$

The aggregated opinions on all indicators are represented as $f_{D_i}(x) = \left\{f^1(x),\ f^2(x),\ \ldots,\ f^h(x),\ h \in D_i,\ i = 1,\ 2,\ \ldots,\ 5\right\}$. Using the weights of individual indicators $\boldsymbol{v} = (v_1,\ v_2,\ \ldots,\ v_h,\ \ldots,\ v_m)$ (calculated by Formula (3)), the opinion aggregation for each dimension is modeled as

$$f_{D_i}(x) = \frac{d}{dx}\left[\left(\sum_{h \in D_i} v_h\left(F^h(x)\right)^{-1}(\rho)\right)^{-1}\right] \tag{17}$$

where D_i is the i th dimension of CMDT.

Following intra-dimension aggregation, the results across all dimensions $f_{CMDT}(x) = \left\{f_{D_1}(x),\ f_{D_2}(x), \ldots f_{D_i}(x), \ldots,\ f_{D_5}(x)\right\}$ are synthesized into a single PDF for overall CMDT maturity:

$$f_{CMDT}(x) = \frac{d}{dx}\left[\left(\sum_{i=1}^{5}\left(F_{D_i}(x)\right)^{-1}(\rho)\right)^{-1}\right] \tag{18}$$

4. Case study

This section presents a case study to demonstrate the application of the proposed CMDT maturity assessment framework. Section 4.1 introduces the background of the target construction machinery. Section 4.2 details the implementation of the framework, from the initial collection of expert judgments to the quantification of overall maturity level.

4.1. Case description

A real-world DT-driven ACS construction project was selected to evaluate the effectiveness of the proposed framework. The ACS represents an advanced form of construction machinery that can accelerate high-rise building construction. However, the operations of the ACS are accompanied by substantial technical challenges, particularly during its ascent alongside the primary building structure. For example, the hydraulic cylinders used for climbing often struggle to maintain synchronized movements owing to factors such as excessive loads and high-speed winds. These challenges increase the risk of instability and, in extreme cases, structural collapse owing to uneven postures during operation.

Stakeholders involved in this project expressed interest in assessing the maturity level of DT implementation. As shown in Fig. 4, the ACS is equipped with a range of sensors designed for both safety and operational monitoring. These include safety-related devices such as inclinometers and leveling instruments, as well as operational sensors such as oil pressure sensors and laser rangefinders. The BIM-based virtual representation of the ACS functions as a centralized platform for data storage, analysis, and simulation. In the presence of potential risks, multi-objective optimization algorithms calculate the optimal operational parameters to balance safety with efficiency. These optimized parameters are subsequently transmitted to the ACS control system, enabling automated adjustments while retaining human oversight. This feedback loop ensures operational optimization and predictive safety measures.

4.2. Assessment results

Questionnaires were distributed to experts directly involved in the ACS DT system, as well as other specialists familiar with the project background. For the latter group, detailed project descriptions and technical information were conveyed through face-to-face communication to ensure informed responses. Of the 17 collected responses, 15 were deemed valid based on their completeness and quality. Two responses were excluded owing to substantial incompleteness and inconsistencies in the expert competency section, suggesting inattentiveness.

The first section of the questionnaire assessed the respondents' expertise in the DT, including relevant certifications or honors. Using the community detection algorithm with a threshold $\theta = 0.7$, the experts were grouped, and each group was assigned a weight based on the opinion group weighting strategy, with $\alpha = 0.7$ and $\beta = 0.3$. The four resulting groups and their weights are summarized in Fig. 5:

Group 1. Respondents 1, 4, 9, and 10, with an assigned importance weight of 0.433.

Group 2. Respondents 2, 6, 8, 11, and 14, with an assigned importance weight of 0.486.

Group 3. Respondents 3, 5, and 15, with an assigned importance weight of 0.531.

Group 4. Respondents 7, 12, and 13, with an assigned importance weight of 0.652.

The second part of the questionnaire focused on scoring each indicator of CMDT. As an example, Fig. 6 shows the opinions of 15 respondents and their fitted PDFs concerning the indicator of operational optimization. The highest point of the PDF indicates the maturity level,

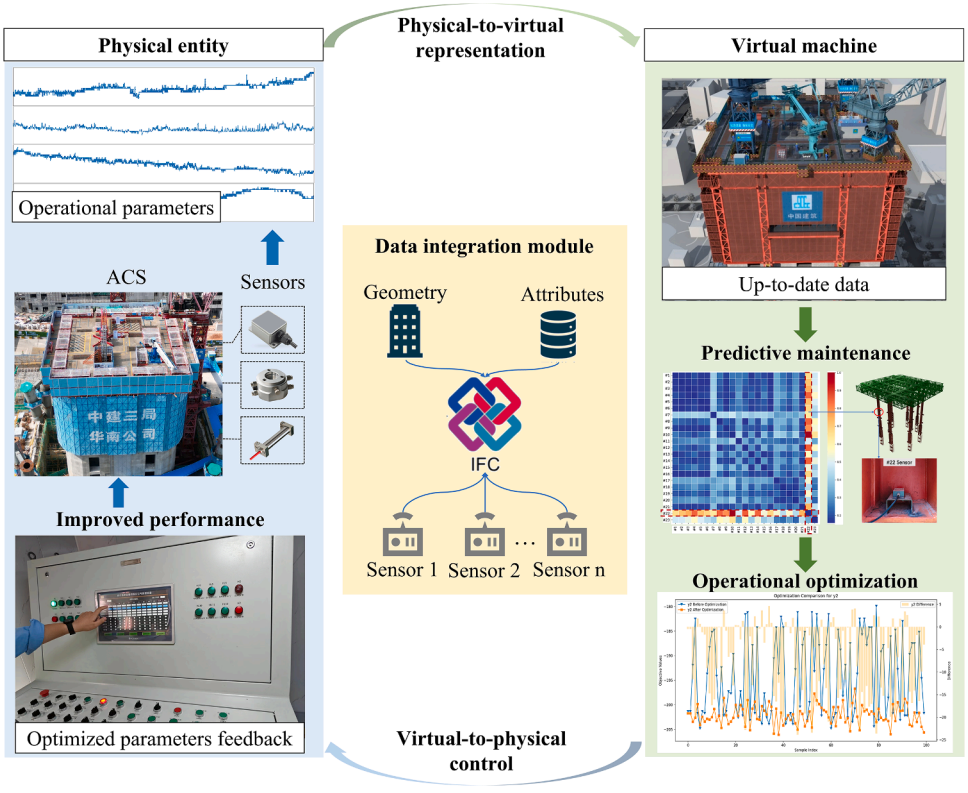


Fig. 4. DT implementation in the ACS.

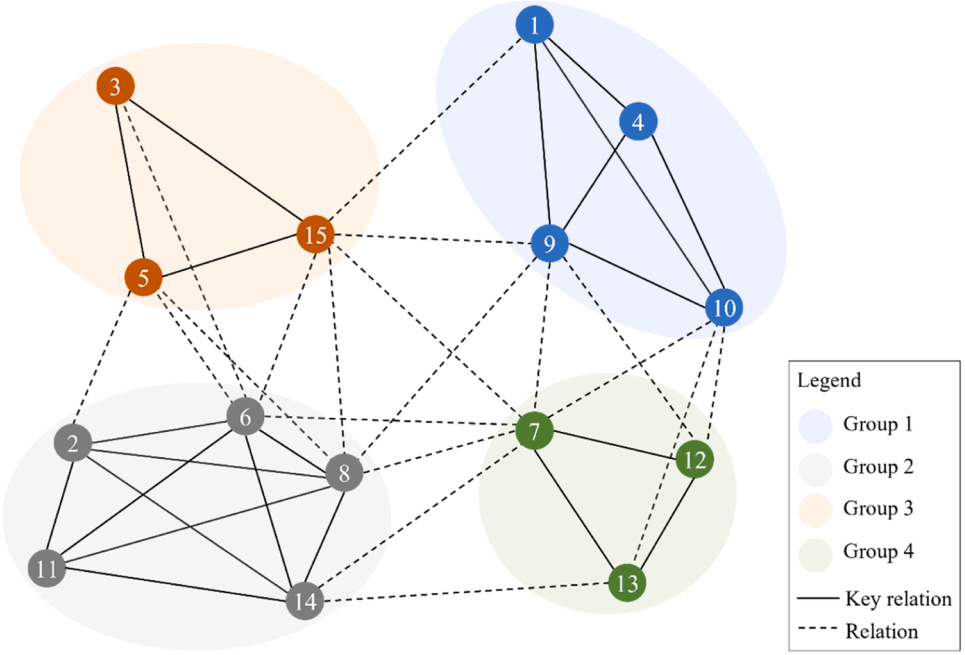


Fig. 5. Opinion grouping and weighting.

and a narrower PDF signifies higher confidence. Among all respondents, the highest reported maturity level was 2.59 (Respondent 13), whereas the lowest was 1.36 (Respondent 8). Notably, Respondent 11′s sharply peaked PDF at Level 3 demonstrates a highly confident judgment, whereas Respondent 2′s flatter curve indicates higher uncertainty. Overall, despite the differences in individual opinions, the evaluation results were predominantly concentrated in Levels 2 and 3.

Based on the grouping results, intra-group opinion aggregation was performed using Formula (14). After extensive testing on the opinion dataset obtained from 15 respondents, λ was set at 0.2, ensuring sufficient overlap of PDF aggregation while preventing the standard deviation range $[\mu-\sigma, \mu+\sigma]$ from spanning three or more maturity levels. Fig. 7 illustrates the optimization process for Group 1 using HO algorithms. The results demonstrate that the objective function converged to a stable

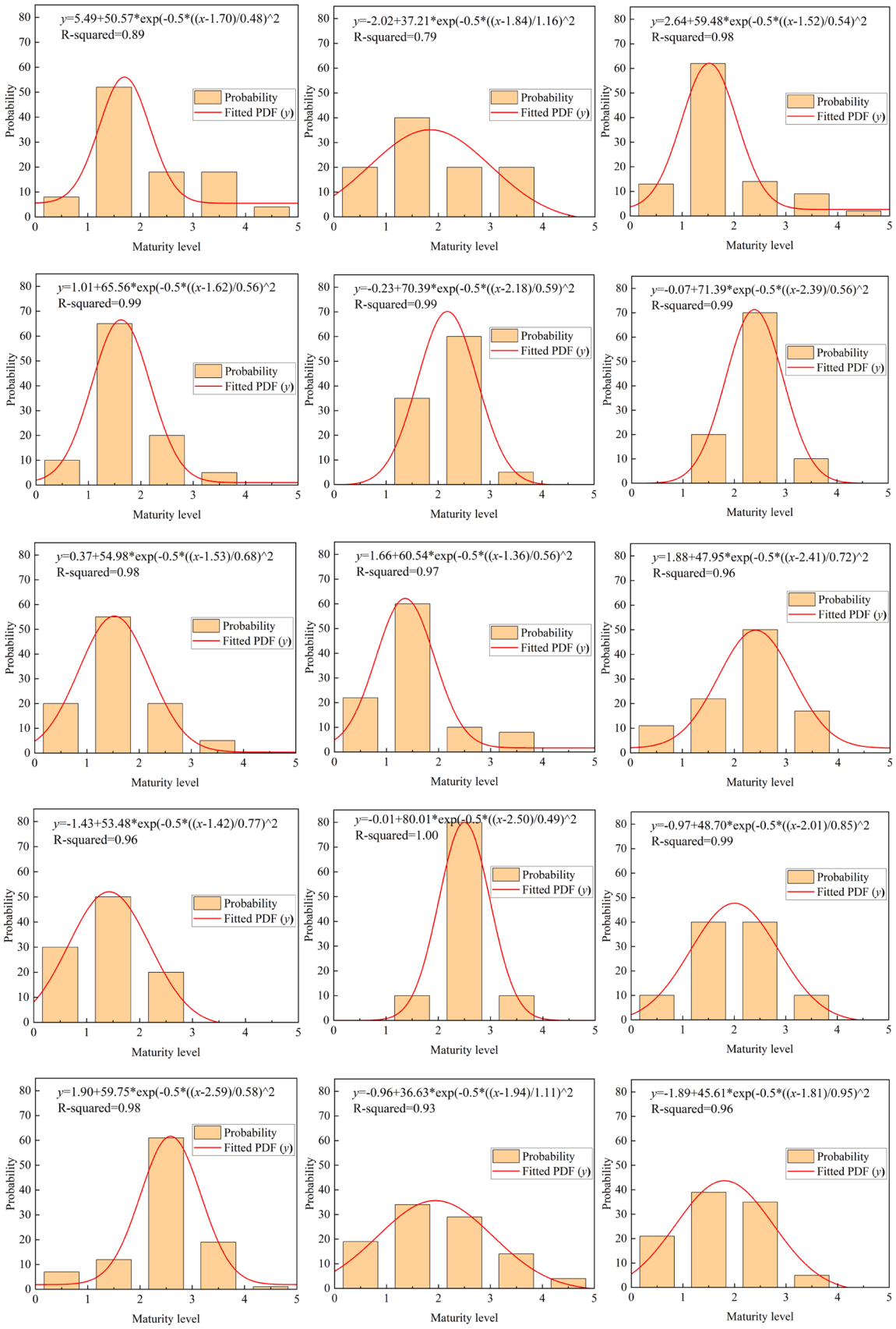


Fig. 6. Fitted PDFs and actual PDFs of 15 respondents concerning the indicator of operational optimization.

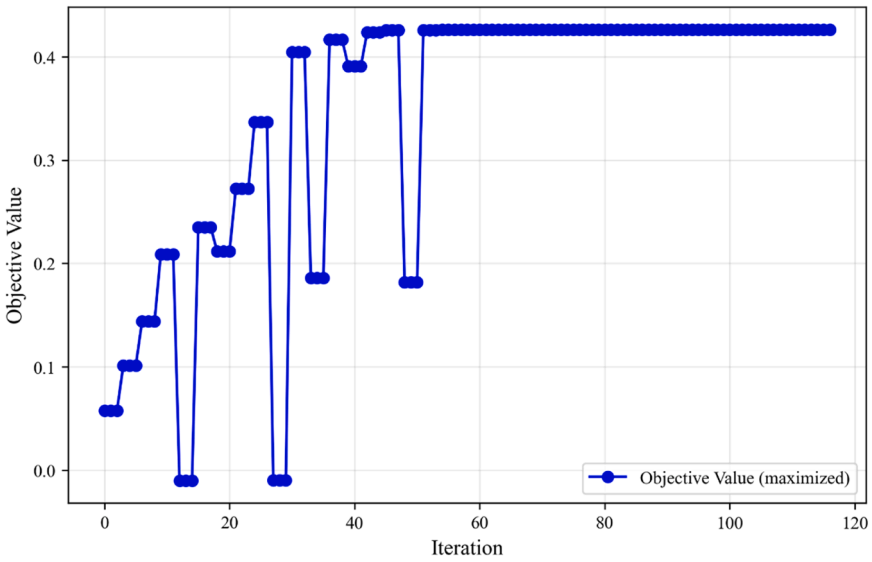


Fig. 7. Convergence of the objective function on opinion aggregation of Group 1.

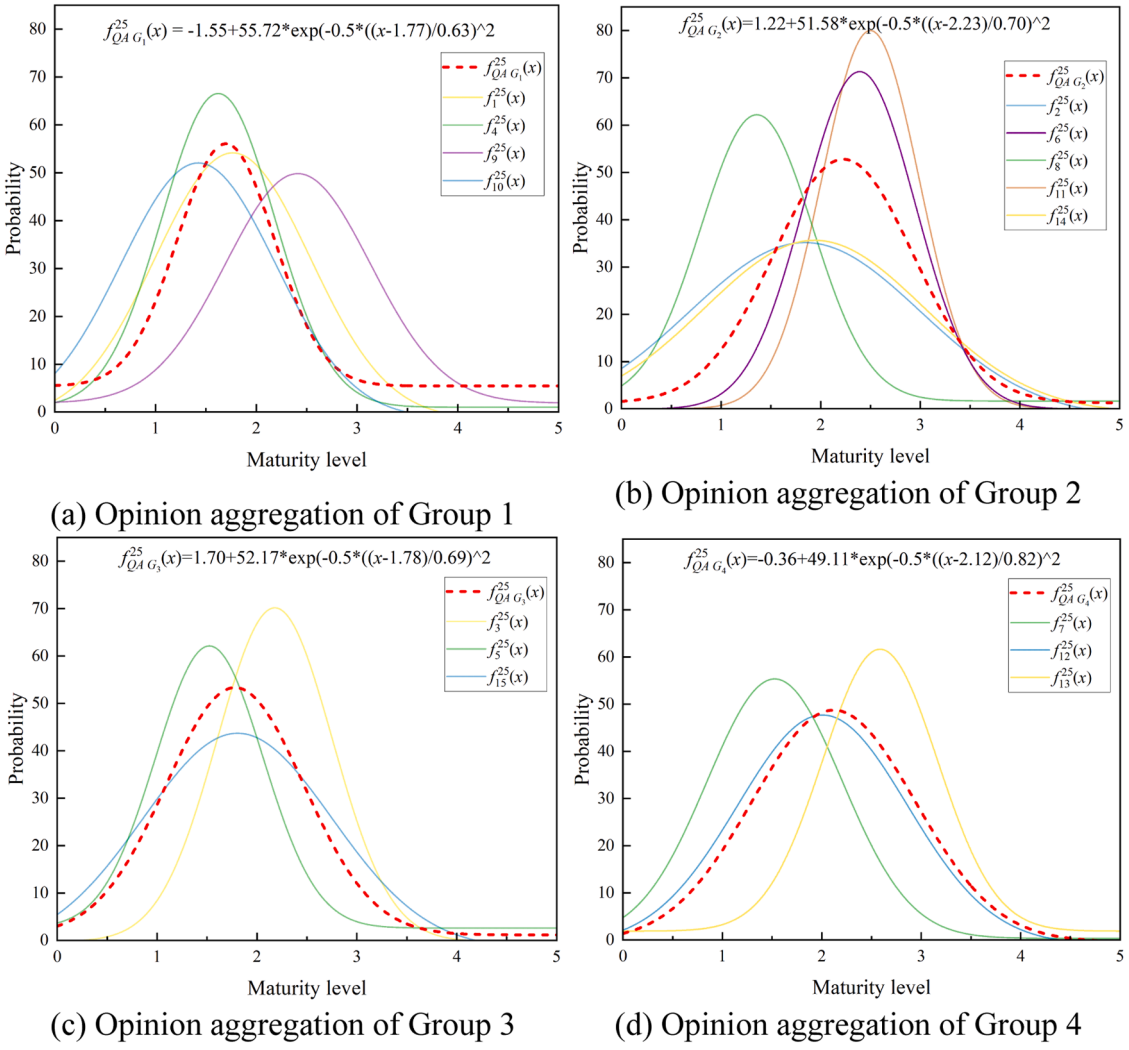


Fig. 8. Intra-group opinion aggregation for individual groups on the indicator of operational optimization.

optimum within approximately 60 iterations. The trajectory of the objective value exhibited an overall upward trend with notable fluctuations during the early stages. Beyond the 60th iteration, the objective value stabilized at approximately 0.43, indicating successful convergence to a near-optimal solution.

Fig. 8 shows the aggregated PDFs for each of the four expert groups. Groups 1 and 3 exhibit a distribution skewed toward Level 2, with peak values centered at approximately 1.8. In contrast, Groups 2 and 4 lean toward Level 3, with peak values at approximately 2.2 and 2.1, respectively. Notably, the PDF of Group is the narrowest among all the groups, indicating a high degree of intra-group consensus and confidence, as evidenced by its low standard deviation of 0.63. Conversely, Group 4 exhibits the widest distribution, with a standard deviation of approximately 0.82, reflecting greater variability in expert opinions and lower internal agreement. Groups 2 and 3 demonstrate intermediate levels of dispersion. Their PDFs are neither sharply peaked nor excessively flat, suggesting a balanced consensus with moderate certainty.

Subsequently, inter-group opinion aggregation for the indicator of operational optimization was performed using Formula (16), yielding an agreed Level 3 maturity, as shown in Fig. 9. According to the maturity levels of indicators listed in Appendix A, this result indicates that the ACS DT system optimizes performance based on real-time sensor data but still requires human intervention for adjustments. Advancing this maturity level necessitates enabling the system to autonomously adapt to changing operational and environmental conditions.

Furthermore, inter-group opinion aggregation was conducted for the remaining 23 indicators, and the results are presented in Table 3. Over 60 % of the indicators were classified as Level 3, approximately 30 % were classified as Level 4, and only two indicators, including real-time data processing and continuous learning, were categorized as Level 2. This suggests that significant improvements are needed in the areas of real-time feedback utilization and dynamic learning to enhance decision-making capabilities.

Finally, collective opinions on individual dimensions and the overall maturity level of the ACS DT system were assessed using Formulas (17) and (18). Fig. 10 shows the aggregated maturity levels across the five dimensions of the CMDT framework. The "Virtual Model" dimension exhibits the highest maturity, with most indicators concentrated around Level 3. This suggests a relatively advanced capability for model fidelity, predictive analytics, and simulation accuracy. The "Communication and Interaction" and "DT Data" dimensions also demonstrate comparatively high maturity levels, reflecting relatively strong performance in data

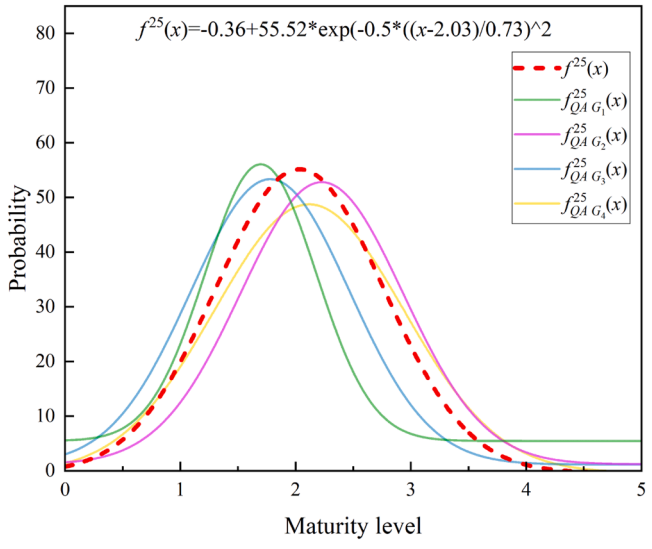


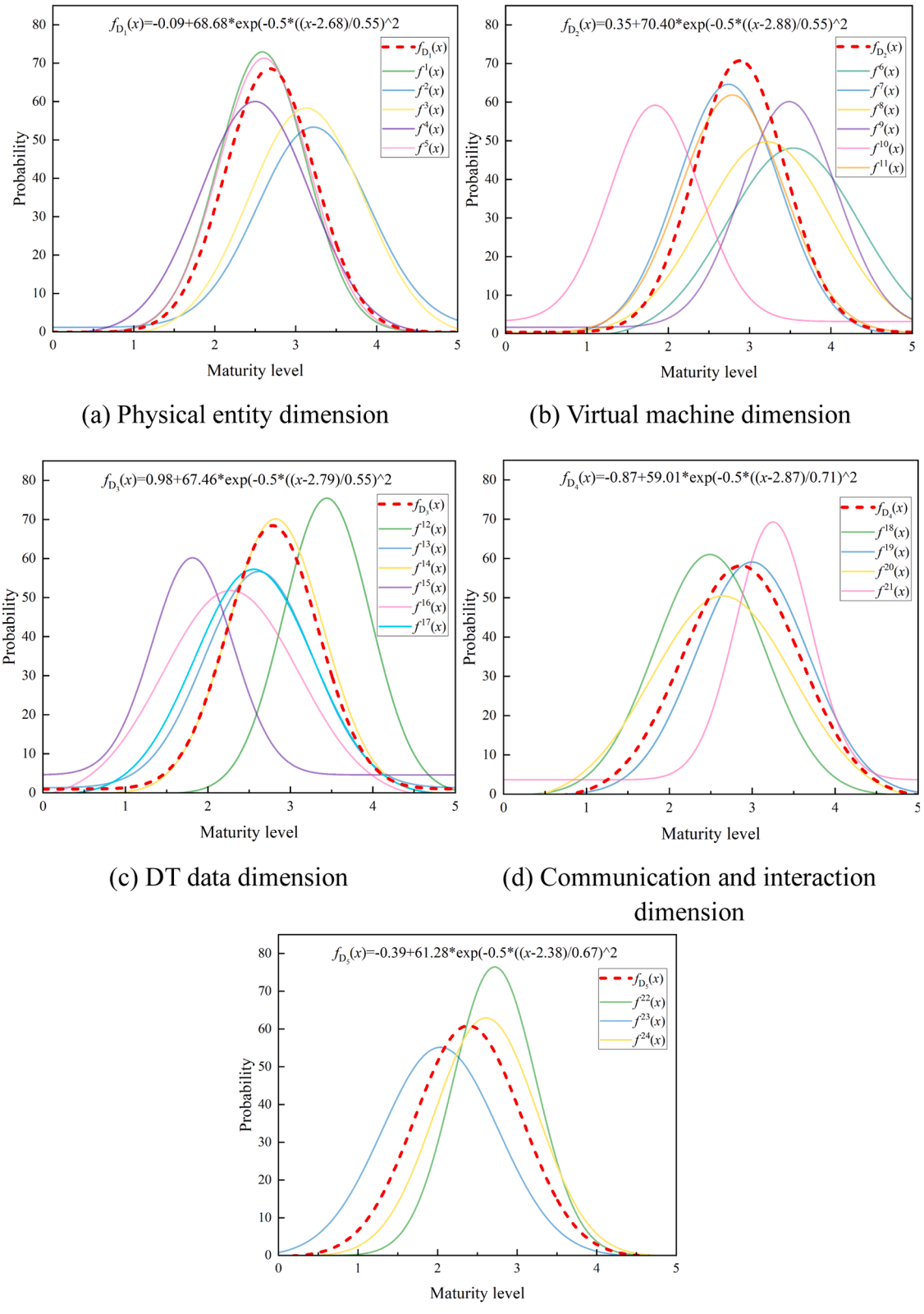
Fig. 9. Inter-group opinion aggregation on the indicator of operational optimization.

Table 3
Opinion aggregation on individual indicators.

Indicator	Mean μ	Standard deviation σ	Fitted PDF $f^h(x)$
Sensor diversity	2.59	0.55	−0.05+72.98*exp(−0.5*((x-2.59)/0.55)^2
Sensor coverage	3.22	0.69	1.17+52.17*exp(−0.5*((x-3.22)/0.69)^2
Data acquisition	3.13	0.71	−1.06+59.36*exp(−0.5*((x-3.13)/0.71)^2
Adaptive actuators	2.50	0.68	−0.40+60.43*exp(−0.5*((x-2.50)/0.68)^2
Level of autonomy	2.61	0.56	−0.07+71.39*exp(−0.5*((x-2.61)/0.56)^2
Model fidelity	3.54	0.82	−1.12+49.25*exp(−0.5*((x-3.54)/0.82)^2
Model update	2.74	0.63	−0.33+64.98*exp(−0.5*((x-2.74)/0.63)^2
Simulation	3.21	0.79	−0.32+50.08*exp(−0.5*((x-3.21)/0.79)^2
Predictive analytics	3.49	0.59	1.67+58.48*exp(−0.5*((x-3.49)/0.59)^2
Continuous learning	1.84	0.55	3.18+56.04*exp(−0.5*((x-1.84)/0.55)^2
Model interoperability	2.79	0.64	0.20+61.65*exp(−0.5*((x-2.79)/0.64)^2
Data completeness	3.44	0.53	−0.23+75.68*exp(−0.5*((x-3.44)/0.53)^2
Data quality	2.62	0.65	1.30+55.37*exp(−0.5*((x-2.62)/0.65)^2
Data standardization	2.82	0.58	−0.23+70.39*exp(−0.5*((x-2.82)/0.58)^2
Real-time data processing	1.81	0.49	4.62+55.56*exp(−0.5*((x-1.81)/0.49)^2
Data accessibility	2.28	0.82	−1.64+53.47*exp(−0.5*((x-2.28)/0.82)^2
Data security	2.56	0.72	−0.64+57.90*exp(−0.5*((x-2.56)/0.72)^2
System connectivity	2.49	0.66	−0.34+61.37*exp(−0.5*((x-2.49)/0.66)^2
Real-time feedback loop	2.99	0.68	−0.24+59.34*exp(−0.5*((x-2.99)/0.68)^2
Cloud integration	2.64	0.85	−1.90+52.38*exp(−0.5*((x-2.64)/0.85)^2
Human–machine interaction	3.25	0.45	3.69+65.58*exp(−0.5*((x-3.25)/0.45)^2
Predictive maintenance	2.72	0.52	−0.06+76.48*exp(−0.5*((x-2.72)/0.52)^2
Operational optimization	2.03	0.73	−0.36+55.52*exp(−0.5*((x-2.03)/0.73)^2
Safety and risk management	2.61	0.64	−0.29+63.22*exp(−0.5*((x-2.61)/0.64)^2

exchange protocols, cloud integration, and structured data processing. In contrast, the "Services" dimension shows the lowest maturity, primarily owing to the underdeveloped functionalities in predictive maintenance and operational optimization. The "Physical Entity" dimension falls in the mid-range, indicating moderate progress in sensor deployment and system autonomy. Overall, these results highlight the necessity of targeted improvements, particularly in enhancing service intelligence and strengthening physical-virtual integration.

Fig. 11 illustrates the overall maturity level of the CMDT implementation, as evaluated across all five dimensions, indicating a predictive and data-driven stage of development, i.e., Level 3. At this level, the ACS DT system can simulate operational performance through real-time sensor input and historical data analysis. However, it still relies on human intervention for adaptive control. To progress toward Level 4, substantial developments are required in autonomous operation, intelligent decision-making, and advanced data management. These enhancements are critical for realizing a fully self-learning DT system capable of end-to-end lifecycle management with minimal human oversight.



(a) Physical entity dimension

(b) Virtual machine dimension

(c) DT data dimension

(d) Communication and interaction dimension

(e) Services dimension

Fig. 10. Intra-dimension opinion aggregation on five dimensions of CMDT.

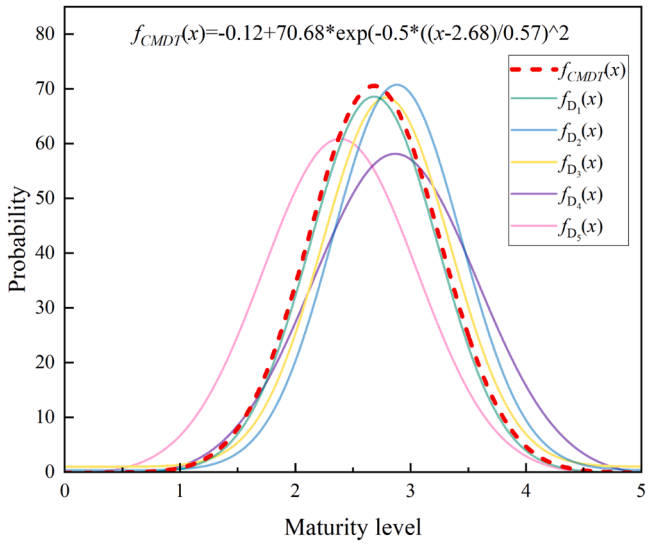


Fig. 11. Overall maturity level of CMDT.

5. Post-assessment validation and discussions

To verify the practical relevance and credibility of the assessment results, a post-assessment validation was conducted using user feedback. Specifically, semi-structured interviews were conducted with key stakeholders involved in the case study project, including three system developers, three system operators, and two project managers. The interviews focused on checking the alignment between the maturity assessment outcomes and users' actual experiences with the ACS DT system. The interviewees consistently corroborated the relatively low maturity score assigned to the "Services" dimension, as well as the comparatively higher maturity levels for the "Virtual Model" and "Communication and Interaction" dimensions. These findings reveal a critical developmental imbalance. Although the DT-driven ACS project has achieved high-fidelity virtual representations, the lack of autonomous "Services" indicates that the current DT system primarily functions as a monitoring platform rather than as a proactive decision-support system. The clustering of results at Level 3 suggests that, while real-time sensor integration and data-driven alerts are operational, the transition to Level 4 is constrained by the persistent need for human intervention within the control loop.

The proposed framework offers several managerial insights. First, it enables evidence-based readiness diagnosis, allowing managers to identify the current CMDT maturity level across multiple dimensions, rather than relying on fragmented technical indicators or subjective judgments. Second, the integration of expert competency-based grouping and weighted opinion aggregation allows managers to align the criticality of DT-supported decisions with the assessed maturity level, thereby enhancing managerial confidence. Third, by revealing specific capability gaps at each maturity level, the framework supports the development of strategic implementation roadmaps. For DT systems at lower maturity levels, the results highlight the need to strengthen foundational capabilities, such as data acquisition, data quality, and system connectivity. For more mature systems, attention can shift toward advanced analytics, closed-loop interaction, and autonomous decision-making.

Of note, the practical adoption of the framework is designed for high usability through an automated pipeline. Complex backend processes for expert grouping and bi-objective optimization can be fully abstracted from end users. Practitioners only need to provide primary survey data, upon which the maturity scores and diagnostic insights can be automatically generated to support strategic improvement planning. Future work may also develop standardized software tools to streamline deployment across diverse construction machinery contexts.

6. Conclusion

This study developed a comprehensive framework for evaluating the CMDT maturity. Based on an extensive literature review and expert consultations, 24 maturity indicators were identified across five dimensions. Maturity levels were systematically defined for each indicator as well as for the overall CMDT maturity model.

To enhance the accuracy and reliability of maturity assessment, this study introduced a opinion grouping and weighting strategy integrated with an improved bi-objective optimization model for aggregating expert opinions. Expert opinions were grouped using community detection algorithms, and weights were assigned to these groups based on both expert competency and the quality of their responses. The bi-objective optimization model was further refined through the introduction of an adjustable penalty weight λ, which balances the overlap of PDFs with the confidence level of the collective opinion. Beyond maturity assessment, the resulting evaluations provide guidance for identifying the evolutionary pathways of CMDTs.

Although the proposed framework provides practical guidance for improving DT solutions in construction machinery, opportunities for further research remain. Future studies should consider incorporating additional criteria to enhance the precision of maturity levels. For example, introducing thresholds for DT model update frequencies can refine the granularity of the assessment. Furthermore, the integration of more objective data, such as sensor logs, can reduce the dependence on expert evaluations. Moreover, extending the application of the framework to other related sectors will enhance its versatility.

CRediT authorship contribution statement

Ruibo Hu: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Wenting Gong:** Visualization, Investigation, Formal analysis. **Ke Chen:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Hanbin Luo:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This study is supported by National Natural Science Foundation of China (72101093, U21A20151) and Science and Technology R&D Program of CSCEC International Operations (CSCI-2023-Z-5). The authors would like to thank the members of iLab at the University of Hong Kong (HKU) for their valuable input on the maturity assessment framework.

Appendix A

Maturity levels of individual CMDT indicators

Dimensions	Indicators	Maturity Levels
Physical entity	Sensor diversity	Level 1: No sensors The machinery is not equipped with any sensors; data collection relies entirely on manual observation and log recording. Level 2: Basic sensors The machinery is equipped with a limited number of sensors, primarily focused on key operations. Level 3: Multiple sensor types The machinery is equipped with multiple types of sensors, supporting various improvements in operations. However, the integration among these sensors remains limited. Level 4: Comprehensive sensor suite The machinery is equipped with a comprehensive sensor suite covering critical operational areas. Data from different types of sensors can be integrated for more detailed analysis. Level 5: Advanced sensor network The machinery is equipped with an advanced sensor network, including various types of sensors. It supports real-time data streaming and advanced analytics.
	Sensor coverage	Level 1: No coverage The machinery is not equipped with any sensors and cannot collect operational performance or health monitoring data. Level 2: Minimal coverage The machinery has a limited number of sensors installed, typically concentrated in one or two key areas. Level 3: Partial coverage The machinery is equipped with sensors across several critical components (e.g., engine, hydraulic system), yet some key areas remain unmonitored. Level 4: Extensive coverage The machinery's sensors cover most critical components, enabling more efficient data collection to improve maintenance and operational decision-making. Level 5: Comprehensive coverage The machinery is equipped with a large number of sensors that monitor all critical and non-critical components, supporting real-time monitoring, predictive analytics, and performance evaluation.
	Data acquisition	Level 1: No real-time data acquisition The machinery lacks the capability for real-time data acquisition. Data must be manually collected, with long intervals between collection cycles. Level 2: Limited real-time data acquisition The machinery provides limited real-time data. Data is recorded at intervals rather than continuously updated. Level 3: Basic real-time monitoring The machinery gathers real-time data from key sensors. Data is transmitted periodically, but the lack of continuous updates restricts the ability to respond quickly to operational changes. Level 4: Continuous real-time data acquisition The machinery continuously collects real-time data from multiple sensors. Data streams uninterrupted to a central system for monitoring and analysis, supporting timely decision-making. Level 5: Advanced real-time acquisition and integration The machinery offers advanced real-time data acquisition, collecting continuous streams from a wide array of sensors. This enables instant insights and autonomous adjustments based on changing conditions.
	Adaptive actuators	Level 1: No adaptive actuators The machinery is equipped with basic actuators that cannot perform any adaptive adjustments. All adjustments must be manually carried out by the operator. Level 2: Basic self-adjustment The machinery includes simple actuators capable of self-adjustments, but the scope of these adjustments is limited. Level 3: Limited adaptive control The actuators can adjust operational parameters based on real-time data from specific sensors. However, operator intervention is still required to achieve optimal performance. Level 4: Semi-autonomous adaptation The machinery is equipped with advanced actuators that can autonomously adjust multiple operational parameters based on a broader set of real-time sensor inputs. Level 5: Fully autonomous adaptive control The machinery features fully autonomous actuators that can adjust all operational parameters in real time without human intervention.
	Level of autonomy	Level 1: Manual operation The machinery is controlled manually with no automation capabilities. Level 2: Assisted operation The machinery includes some automation features, such as basic alerts or notifications, but the operator retains full control over all functions and decisions. Level 3: Semi-autonomous operation The machinery can autonomously perform specific tasks based on predefined parameters, but operator supervision and intervention are required. Level 4: Highly autonomous operation The machinery can operate independently for extended periods with minimal human intervention. It can adapt to changing conditions, though operator oversight is still recommended. Level 5: Fully autonomous operation The machinery operates entirely autonomously, capable of making complex decisions and adjustments.
Virtual model	Model fidelity	Level 1: Basic representation The virtual model consists of simple 2D CAD drawings or basic geometric shapes. Level 2: Static 3D representation A static 3D model of the machinery is created, providing a basic visual representation but lacking detailed functional

(*continued on next page*)

(*continued*)

		characteristics. Level 3: Detailed 3D model The virtual model features a detailed 3D representation, accurately reflecting the components and structure of the physical machinery. Some functional aspects are included, but real-time data integration is absent. Level 4: Dynamic and realistic model The virtual model integrates real-time data to accurately reflect the operational state of the machinery. Level 5: High-fidelity autonomous model The virtual model is a high-fidelity digital twin that can precisely replicate the physical machinery. It continuously learns and adapts based on real-time data.
	Model update	Level 1: No updates The virtual model is static and remains unchanged after its initial creation. Level 2: Manual updates The virtual model is manually updated at irregular intervals. Level 3: Periodic updates The virtual model is updated on a scheduled basis to reflect changes in the physical machinery. Level 4: Semi-automatic updates The virtual model incorporates a semi-automated update process where certain changes are automatically captured, while others still require manual intervention. Level 5: Continuous real-time updates The virtual model achieves continuous real-time updates. All changes can be synchronized immediately without human intervention.
	Simulation	Level 1: No simulation capability The virtual model does not support any simulation functionality. Level 2: Basic scenario simulation The virtual model can simulate simple scenarios based on static data inputs. Level 3: Dynamic simulation The virtual model can perform dynamic simulations incorporating real-time data. Level 4: Advanced predictive simulation The virtual model supports advanced predictive simulations, using historical and real-time data to forecast outcomes and optimize machinery performance in complex scenarios. Level 5: Autonomous simulation and optimization The virtual model has fully autonomous simulation capabilities, running real-time simulations to optimize performance. It adjusts operational parameters automatically based on continuous analysis.
	Predictive analytics	Level 1: No predictive analytics The virtual model has no predictive analytics capabilities. Level 2: Basic predictive analytics The virtual model provides basic predictions based on limited historical data, without incorporating real-time data or complex variables. Level 3: Intermediate predictive analytics The virtual model combines historical data with some real-time inputs to generate predictions, but the analysis does not account for all relevant factors. Level 4: Advanced predictive modeling The virtual model uses advanced predictive analytics techniques, such as deep learning, to analyze comprehensive datasets and provide detailed, accurate predictions. Level 5: Autonomous predictive analytics The virtual model possesses fully autonomous predictive analytics capabilities, continuously learning from real-time data and operational outcomes to adjust predictions dynamically based on changing conditions.
	Continuous learning	Level 1: No learning capability The virtual model operates on predefined rules. There is no mechanism to integrate new data, feedback, or changing conditions. Level 2: Basic data integration The virtual model periodically integrates new data to support model refinement, but updates are manual and infrequent. Level 3: Automated data integration The virtual model automatically integrates new data and updates optimization periodically. However, learning is limited to predefined parameters, with no advanced self-optimization. Level 4: Feedback-driven adaptive learning The virtual model actively learns from real-time data and feedback, dynamically improving its model and decision-making performance. Level 5: Self-learning and optimization The virtual model has fully autonomous, continuous learning capabilities. It evolves based on external factors in complex scenarios and actively seeks performance improvements.
	Model interoperability	Level 1: No interoperability The virtual model operates independently and cannot integrate with other systems or models. There are no data exchange or collaboration mechanisms. Level 2: Limited interoperability The virtual model can connect with a few specific systems or models, but the integration is manual and limited. Data exchange is infrequent and lacks standardization. Level 3: Basic interoperability The virtual model supports basic integration with multiple systems and models using standardized data formats. Level 4: Enhanced interoperability The virtual model integrates with a wide range of systems and models across different domains, supporting multi-scale and multi-physics interactions. Level 5: Full interoperability The virtual model offers full interoperability, seamlessly integrating with various systems and models, and supporting extensive multi-scale, multi-physics, and multi-level expansion.
DT data	Data completeness	Level 1: Incomplete data The dataset has significant gaps, with many necessary data points missing. Level 2: Partially complete data

(*continued on next page*)

(*continued*)

		Some necessary data is available, but there are still significant gaps. Level 3: Mostly complete data Most of the necessary data is collected, but certain key elements are still missing, affecting the accuracy and reliability of data analysis. Level 4: Highly complete data The dataset is mostly complete, with only minor gaps, ensuring reliable data analysis. Level 5: Complete data The dataset is complete. There are no gaps, supporting comprehensive, high-quality data analysis.
	Data quality	Level 1: Poor quality Data is highly inaccurate, inconsistent, and unreliable. Frequent errors severely impact the effectiveness of data analysis. Level 2: Fair quality Data contains many inaccuracies and inconsistencies. Data quality issues limit the effectiveness of analysis and decision-making. Level 3: Moderate quality Data is generally reliable, but there are some errors and inconsistencies. While it can be used for analysis, it may affect the credibility of the results. Level 4: High quality Data is mostly accurate and consistent, with only minor errors, providing reliable analysis results. Level 5: Excellent quality Data is accurate, complete, and reliable, supporting precise and detailed analysis results.
	Data standardization	Level 1: No standardization Data is stored in various formats and structures, with no consistent definitions, leading to confusion, inconsistency, and difficulties in data integration and analysis. Level 2: Preliminary standardization Some data has been standardized, but many inconsistencies remain. Data formats differ across systems, limiting integration and reuse. Level 3: Basic standardization Key data has common definitions and formats, but some data still lacks consistency, affecting integration and analysis. Level 4: Advanced standardization Most data is standardized. Definitions are clear and formats are consistent, facilitating better integration and analysis. Level 5: Fully standardized All data is fully standardized, with comprehensive definitions and consistent formats, supporting seamless integration, interoperability, and efficient analysis.
	Real-time data processing	Level 1: No real-time processing Data is processed manually or in batch mode, with significant delays. Level 2: Occasional real-time processing Some data is processed in real time, but key data still experiences delayed processing, hindering timely responses. Level 3: Limited real-time processing Key data is processed in real time, but processing scope and frequency are limited. Non-critical data is still processed with delays. Level 4: Comprehensive real-time processing All critical data is processed in real time, supporting timely responses. Only a small portion of non-critical data is not processed in real time. Level 5: Advanced real-time processing All data is processed instantly, with real-time capture and analysis of all dynamic changes, enabling fast and accurate decision-making.
	Data accessibility	Level 1: Limited access Data is stored on individual machines or devices, with only a few stakeholders able to retrieve it, typically requiring manual operations. Level 2: Basic centralized access Data is stored centrally in a central repository, accessible only to authorized personnel through basic interfaces, with no support for real-time retrieval. Level 3: Role-based access Data is accessed in real time through a centralized system, with role-based permissions ensuring that different stakeholders can access the appropriate data. Level 4: Widespread access Data is broadly accessible across various devices and platforms, with stakeholders at all levels accessing data relevant to their roles. Level 5: Smart access Data access is fully context-aware, providing the correct data to the right users at the right time.
	Data security	Level 1: No security measures No data security measures are implemented, posing significant risks to confidentiality and integrity. Level 2: Basic security protocols Basic security measures are implemented, such as password protection and limited access control. Level 3: Standard security practices Standard security practices are implemented, including encryption, regular access audits, and basic intrusion detection. Level 4: Advanced security framework A comprehensive security framework is in place, combining advanced measures such as multi-factor authentication, continuous monitoring, and incident response plans. Level 5: Robust and adaptive security A robust and adaptive security strategy is employed, featuring proactive threat detection, real-time monitoring, and ongoing improvements to address emerging threats.
Connection and interaction	System connectivity	Level 1: No connectivity The physical machinery operates independently without any connection to digital systems, cloud platforms, or other machines. Level 2: Local connectivity The physical machinery is connected to a local network, allowing communication with nearby management systems. Level 3: Cloud connectivity The physical machinery is connected to a cloud platform, enabling remote monitoring, control, and data access. Level 4: Integrated system connectivity

(*continued on next page*)

(*continued*)

		The physical machinery is part of a digital twin system, interacting with the virtual model. This bidirectional connection allows the machinery to exchange data with external systems. Level 5: Fully connected The physical machinery is fully connected to an autonomous, real-time ecosystem. The virtual model seamlessly exchanges data with the physical machinery.
	Real-time feedback loop	Level 1: No feedback loop There is no feedback loop. Level 2: Limited feedback Basic feedback mechanisms are in place, but the feedback loop is not real-time. Level 3: Basic real-time feedback The virtual model can send real-time feedback to the physical machinery, but the adjustments are limited to simple changes based on predefined parameters. Level 4: Advanced real-time feedback A comprehensive real-time feedback loop is established between the model and the machinery, allowing the physical machinery to make dynamic adjustments based on real-time data and operational conditions. Level 5: Fully autonomous feedback loop The digital twin system autonomously controls the machinery in real time, continuously adjusting and optimizing machine performance based on predictive analysis and external data inputs.
	Cloud integration	Level 1: No cloud integration The machinery operates independently, with no connection to the cloud. Level 2: Limited cloud integration Data from the machinery can be manually uploaded to the cloud for storage or analysis, but this process is not continuous or automated. Level 3: Basic cloud integration The machinery is connected to the cloud, supporting real-time data transmission and remote monitoring. Level 4: Advanced cloud integration Real-time data streaming from the machinery to the cloud is established, enabling continuous monitoring, control, and analysis from any location. Level 5: Cloud integration ecosystem The machinery is fully integrated into a cloud-based ecosystem, supporting seamless interaction between physical machinery, virtual model, and other systems such as supply chains and analytics tools.
	Human-machine interaction	Level 1: Manual interaction All interactions with the machinery are manual, with no digital interaction. The operator must manage the machine's operations on-site. Level 2: Basic digital interface The machine includes a basic digital interface, such as a simple control panel or screen, allowing the operator to remotely interact with certain parts of the machinery. Level 3: Remote digital interaction The operator can interact with the machine remotely through a digital platform, using a centralized system to control machine operations and monitor performance. Level 4: Advanced human-machine interaction The system integrates advanced interaction technologies, such as AR/VR, enabling operators to make decisions in an immersive digital environment for greater control precision. Level 5: Immersive autonomous interaction The system provides fully immersive human-machine interaction through AR/VR interfaces, allowing operators to supervise and control machine operations in a virtual environment. Human input is focused more on strategic decisions, while most operational tasks are autonomously managed by the digital twin system.
Services	Predictive maintenance	Level 1: Reactive maintenance Maintenance is entirely reactive, occurring only after a failure or damage has occurred. Level 2: Basic condition monitoring Maintenance is still primarily reactive, but status data provides basic insights into machinery health. Level 3: Data-driven alerts Alerts are generated based on historical trends and sensor data, allowing for intervention before failures occur. Level 4: Advanced predictive analytics The system not only predicts failures but also forecasts optimal maintenance schedules. Level 5: Autonomous predictive maintenance Maintenance is fully autonomous, with the system automatically scheduling and executing maintenance tasks based on real-time data and self-learning algorithms, requiring no human intervention.
	Operational optimization	Level 1: No optimization Adjustments are made manually based on the operator's experience. Level 2: Basic manual adjustments Basic operational adjustments are made based on real-time data, improving the operator's optimization efficiency. Level 3: Data-driven optimization The system provides optimization recommendations based on real-time data. Level 4: Automated operational adjustments The system automatically optimizes real-time operational parameters, adjusting machine performance based on environmental and operational factors. Level 5: Fully autonomous and continuous optimization The machine autonomously and continuously optimizes all aspects of its operation based on real-time data, historical trends, and predictive analysis. Over time, it self-learns to maximize efficiency.
	Safety and risk management	Level 1: No safety monitoring There are no digital safety measures or monitoring systems in place, and safety practices rely entirely on manual operations, operator awareness, and experience. Level 2: Basic safety monitoring Basic safety monitoring systems are in place, such as simple alerts for key safety scenarios (e.g., machinery overheating or operator alarms), with a focus on reactions after issues occur. Level 3: Real-time risk assessment

(*continued on next page*)

(*continued*)

	The system performs real-time risk assessments using sensor data to identify potential hazards. It generates alerts and notifications to raise operator safety awareness and speed up responses to safety issues. Level 4: Proactive safety response Based on historical data and real-time inputs, potential risks are predicted, and proactive safety measures are implemented. Automatic reporting tools track compliance and performance metrics, improving the efficiency of safety management. Level 5: Autonomous safety management The safety management system is fully autonomous, continuously assessing risks and implementing safety measures without human intervention. The safety system integrates with other systems, adapting in real time to changing conditions.

Data availability

The authors do not have permission to share data.

References

[1] H. Xie, M. Xin, C. Lu, J. Xu, Knowledge map and forecast of digital twin in the construction industry: state-of-the-art review using scientometric analysis, J. Clean. Prod. 383 (2023) 135231, https://doi.org/10.1016/j.jclepro.2022.135231.

[2] R. Hu, K. Chen, W. Jiang, H. Luo, IFC data extension for real-time safety monitoring of automated construction in high-rise building projects, Autom. Constr. 162 (2024) 105408, https://doi.org/10.1016/j.autcon.2024.105408.

[3] W. Jiang, Y. Liu, K. Chen, Y. Liu, L. Ding, Early-warning of unsafe hoisting operations: an integration of digital twin and knowledge graph, Dev. Built Environ. 19 (2024) 100490, https://doi.org/10.1016/j.dibe.2024.100490.

[4] F. Jiang, L. Ma, T. Broyd, K. Chen, Digital twin and its implementations in the civil engineering sector, Autom. Constr. 130 (2021) 103838, https://doi.org/10.1016/j.autcon.2021.103838.

[5] F. Tao, H. Zhang, C. Zhang, Advancements and challenges of digital twins in industry, Nat. Comput. Sci. 4 (2024) 169–177, https://doi.org/10.1038/s43588-024-00603-w.

[6] R. Sacks, I. Brilakis, E. Pikas, H.S. Xie, M. Girolami, Construction with digital twin information systems, Data-Centr. Eng. 1 (2020) e14, https://doi.org/10.1017/dce.2020.16.

[7] D.-G.J. Opoku, S. Perera, R. Osei-Kyei, M. Rashidi, Digital twin application in the construction industry: a literature review, J. Build. Eng. 40 (2021) 102726, https://doi.org/10.1016/j.jobe.2021.102726.

[8] S. Su, R.Y. Zhong, Y. Jiang, J. Song, Y. Fu, H. Cao, Digital twin and its potential applications in construction industry: state-of-art review and a conceptual framework, Adv. Eng. Inform. 57 (2023) 102030, https://doi.org/10.1016/j.aei.2023.102030.

[9] V.V. Tuhaise, J.H.M. Tah, F.H. Abanda, Technologies for digital twin applications in construction, Autom. Constr. 152 (2023) 104931, https://doi.org/10.1016/j.autcon.2023.104931.

[10] H. Noh, L. Mortara, S. Lee, Development of a maturity model for technology intelligence, Technol. Forecast. Soc. Change 192 (2023) 122525, https://doi.org/10.1016/j.techfore.2023.122525.

[11] J.-F. Uhlenkamp, J.B. Hauge, E. Broda, M. Lutjen, M. Freitag, K.-D. Thoben, Digital twins: a maturity model for their classification and evaluation, IEEE Access 10 (2022) 69605–69635, https://doi.org/10.1109/ACCESS.2022.3186353.

[12] Y. Liu, J. Feng, J. Lu, S. Zhou, A review of digital twin capabilities, technologies, and applications based on the maturity model, Adv. Eng. Inform. 62 (2024) 102592, https://doi.org/10.1016/j.aei.2024.102592.

[13] W. Hu, J. Fang, T. Zhang, Z. Liu, J. Tan, A new quantitative digital twin maturity model for high-end equipment, J. Manuf. Syst. 66 (2023) 248–259, https://doi.org/10.1016/j.jmsy.2022.12.012.

[14] Z.S. Chen, K.D. Chen, Y.Q. Xu, W. Pedrycz, M.J. Skibniewski, Multiobjective optimization-based decision support for building digital twin maturity measurement, Adv. Eng. Inform. 59 (2024) 102245, https://doi.org/10.1016/j.aei.2023.102245.

[15] T. Li, Y. Rui, S. Zhao, Y. Zhang, H. Zhu, X. Li, A quantitative digital twin maturity model for underground infrastructure based on D-ANP, Tunnell. Undergr. Space Technol. 146 (2024), https://doi.org/10.1016/j.tust.2024.105612.

[16] T. Albert, M.G. Moehrle, S. Meyer, Technology maturity assessment based on blog analysis, Technol. Forecast. Soc. Change 92 (2015) 196–209, https://doi.org/10.1016/j.techfore.2014.08.011.

[17] F. Tao, F. Sui, A. Liu, Q. Qi, M. Zhang, B. Song, Z. Guo, S.C.-Y. Lu, A.Y.C. Nee, Digital twin-driven product design framework, Int. J. Prod. Res. 57 (2019) 3935–3953, https://doi.org/10.1080/00207543.2018.1443229.

[18] T.H.-J. Uhlemann, C. Schock, C. Lehmann, S. Freiberger, R. Steinhilper, The digital twin: demonstrating the potential of real time data acquisition in production systems, Procedia Manuf. 9 (2017) 113–120, https://doi.org/10.1016/j.promfg.2017.04.043.

[19] D. Jones, C. Snider, A. Nassehi, J. Yon, B. Hicks, Characterising the digital twin: a systematic literature review, CIRP J. Manuf. Sci. Technol. 29 (2020) 36–52, https://doi.org/10.1016/j.cirpj.2020.02.002.

[20] M. Grieves, J. Vickers, Digital twin: mitigating unpredictable, undesirable emergent behavior in complex systems. Transdisciplinary Perspectives on Complex Systems, Springer International Publishing, Cham, 2017, pp. 85–113, https://doi.org/10.1007/978-3-319-38756-7_4.

[21] W. Kritzinger, M. Karner, G. Traar, J. Henjes, W. Sihn, Digital twin in manufacturing: a categorical literature review and classification, IFAC-PapersOnLine 51 (2018) 1016–1022, https://doi.org/10.1016/j.ifacol.2018.08.474.

[22] R. Klar, N. Arvidsson, V. Angelakis, Digital twins' maturity: the need for interoperability, IEEE Syst. J. 18 (2024) 713–724, https://doi.org/10.1109/JSYST.2023.3340422.

[23] F.G. Medina, A.W. Umpierrez, V. Martinez, H. Fromm, A maturity model for digital twin implementations in the commercial aerospace OEM industry, 2021, in: 10th International Conference on Industrial Technology and Management (ICITM), IEEE, 2021, pp. 149–156, https://doi.org/10.1109/ICITM52822.2021.00034.

[24] L.F. Villegas, M. Macchi, A. Polenghi, Towards a maturity model for intelligent digital twins in manufacturing, in: M. Thürer, R. Riedel, G. von Cieminski, D. Romero (Eds.), IFIP International Conference on Advances in Production Management Systems, Springer Nature Switzerland, Cham, 2024, pp. 293–306, https://doi.org/10.1007/978-3-031-71633-1_21.

[25] X. Liu, G. Li, F. Xiang, B. Tao, G. Jiang, Expert opinion aggregation-based decision support for human-robot collaboration digital twin maturity assessment, J. Ind. Inf. Integr. 42 (2024) 100710, https://doi.org/10.1016/j.jii.2024.100710.

[26] Y. Maïzi, A. Arcand, Y. Bendavid, Digital twin in healthcare: classification and typology of models based on hierarchy, application, and maturity, Internet Things 28 (2024) 101379, https://doi.org/10.1016/j.iot.2024.101379.

[27] L. Chen, X. Xie, Q. Lu, A.K. Parlikad, M. Pitt, J. Yang, Gemini principles-based digital twin maturity model for asset management, Sustainability 13 (2021) 1–15, https://doi.org/10.3390/su13158224.

[28] A. Schumacher, S. Erol, W. Sihn, A maturity model for assessing industry 4.0 readiness and maturity of manufacturing enterprises, Procedia CIRP 52 (2016) 161–166, https://doi.org/10.1016/j.procir.2016.07.040.

[29] T. Gollhardt, S. Halsbenning, A. Hermann, A. Karsakova, J. Becker, Development of a digital transformation maturity model for IT companies, in: 2020 IEEE 22nd Conference on Business Informatics (CBI), IEEE, 2020, pp. 94–103, https://doi.org/10.1109/CBI49978.2020.00018.

[30] B. Wernicke, L. Stehn, A.A. Sezer, M. Thunberg, Introduction of a digital maturity assessment framework for construction site operations, Int. J. Construct. Manag. 23 (2023) 898–908, https://doi.org/10.1080/15623599.2021.1943629.

[31] Z.-S. Chen, Z. Zhu, X.-J. Wang, F. Chiclana, E. Herrera-Viedma, M.J. Skibniewski, Multiobjective optimization-based collective opinion generation with fairness concern, IEEE Trans. Syst. Man Cybern. 53 (2023) 5729–5741, https://doi.org/10.1109/TSMC.2023.3273715.

[32] S. Bhandari, M.R. Hallowell, Identifying and controlling biases in expert-opinion research: guidelines for variations of Delphi, nominal group technique, and focus groups, J. Manag. Eng. 37 (2021) 4021015, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000909.

[33] Z.S. Chen, M.D. Zhou, K.S. Chin, A. Darko, X.J. Wang, W. Pedrycz, Optimized decision support for BIM maturity assessment, Autom. Constr. 149 (2023), https://doi.org/10.1016/j.autcon.2023.104808.

[34] J.R. Trillo, E. Herrera-Viedma, J.A. Morente-Molinera, F.J. Cabrerizo, A large scale group decision making system based on sentiment analysis cluster, Inf. Fusion 91 (2023) 633–643, https://doi.org/10.1016/j.inffus.2022.11.009.

[35] G.J. Oyewole, G.A. Thopil, Data clustering: application and trends, Artif. Intell. Rev. 56 (2023) 6439–6475, https://doi.org/10.1007/s10462-022-10325-y.

[36] H. Silber, D. Danner, B. Rammstedt, The impact of respondent attentiveness on reliability and validity, Int. J. Soc. Res. Methodol. 22 (2019) 153–164, https://doi.org/10.1080/13645579.2018.1507378.

[37] J. Fortin, D. Dubois, H. Fargier, Gradual numbers and their application to fuzzy interval analysis, IEEE Trans. Fuzzy Syst. 16 (2008) 388–402, https://doi.org/10.1109/TFUZZ.2006.890680.

[38] L. Hortovanyi, R.E. Morgan, I.V. Herceg, D. Djuricin, R. Hanak, D. Horvath, M. L. Mocan, A. Romanova, R.Z. Szabo, Assessment of digital maturity: the role of resources and capabilities in digital transformation in B2B firms, Int. J. Prod. Res. 61 (2023) 8043–8061, https://doi.org/10.1080/00207543.2022.2164087.

[39] C. Ji, X. Lu, W. Zhang, A biobjective optimization model for expert opinions aggregation and its application in group decision making, IEEE Syst. J. 15 (2021) 2834–2844, https://doi.org/10.1109/JSYST.2020.3027716.

[40] X. Liu, A. Ghorpade, Y.L. Tu, W.J. Zhang, A novel approach to probability distribution aggregation, Inf. Sci. 188 (2012) 269–275, https://doi.org/10.1016/j.ins.2011.11.002.

[41] M. Cai, Y. Lin, B. Han, C. Liu, W. Zhang, On a simple and efficient approach to probability distribution function aggregation, IEEE Trans. Syst. Man Cybern. 47 (2016) 1–100, https://doi.org/10.1109/TSMC.2016.2531647.
[42] F. Tao, C. Zhang, Q. Qi, H. Zhang, Digital twin maturity model, Comput. Integrat. Manuf. Syst. 28 (2022) 1267–1281, https://doi.org/10.13196/j.cims.2022.05.001.
[43] G. Korovin, Digital twins in the industry: maturity, functions, effects, in: Lecture Notes in Information Systems and Organisation, 2022, pp. 1–12, https://doi.org/10.1007/978-3-030-94617-3_1.
[44] R. Klar, A. Fredriksson, V. Angelakis, Assessing the maturity of digital twinning solutions for ports, in: 2023 IEEE International Conference on Pervasive Computing and Communications Workshops and Other Affiliated Events, PerCom Workshops 2023, 2023, pp. 552–557, https://doi.org/10.1109/PerComWorkshops56833.2023.10150378.
[45] D. Schabany, T.H. Hülsmann, A. Schmetz, Development of a maturity assessment model for digital twins in battery cell industry, Procedia CIRP. 120 (2023) 946–951, https://doi.org/10.1016/j.procir.2023.09.105.
[46] A. Thaduri, Digital Twin: Definitions, Classification, and Maturity, Lecture Notes in Mechanical Engineering, 2024, pp. 585–599, https://doi.org/10.1007/978-3-031-39619-9_43.
[47] J.F. Uhlenkamp, J.B. Hauge, E. Broda, M. Lutjen, M. Freitag, K.D. Thoben, Digital twins: a maturity model for their classification and evaluation, IEEE Access 10 (2022) 69605–69635, https://doi.org/10.1109/ACCESS.2022.3186353.
[48] M. Haraguchi, T. Funahashi, F. Biljecki, Assessing governance implications of city digital twin technology: a maturity model approach, Technol. Forecast. Soc. Change 204 (2024), https://doi.org/10.1016/j.techfore.2024.123409.
[49] R. Manickam, J. Vollmar, G.M. Prabhakar, User experience–digital twin maturity model (UX-DTMM), in: Smart Innovation, Systems and Technologies, 2023, pp. 877–889, https://doi.org/10.1007/978-981-99-0264-4_72.
[50] Y. Liu, J. Feng, J. Lu, S. Zhou, A review of digital twin capabilities, technologies, and applications based on the maturity model, Adv. Eng. Inform. 62 (2024), https://doi.org/10.1016/j.aei.2024.102592.
[51] C. Yang, X. Tu, J. Autiosalo, R. Ala-Laurinaho, J. Mattila, P. Salminen, K. Tammi, Extended reality application framework for a digital-twin-based smart crane, Appl. Sci. (2022) 12, https://doi.org/10.3390/app12126030.
[52] M. Niederberger, S. Köberich, Coming to consensus: the Delphi technique, Eur. J. Cardiovasc. Nurs. 20 (2021) 692–695, https://doi.org/10.1093/eurjcn/zvab059.
[53] R.M. Alvarez, L.R. Atkeson, I. Levin, Y. Li, Paying attention to inattentive survey respondents, Polit. Anal. 27 (2019) 145–162, https://doi.org/10.1017/pan.2018.57.
[54] K.C. Lichtendahl, Y. Grushka-Cockayne, R.L. Winkler, Is it better to average probabilities or quantiles? Manage Sci. 59 (2013) 1594–1611, https://doi.org/10.1287/mnsc.1120.1667.
[55] M.H. Amiri, N. Mehrabi Hashjin, M. Montazeri, S. Mirjalili, N. Khodadadi, Hippopotamus optimization algorithm: a novel nature-inspired optimization algorithm, Sci. Rep. 14 (2024) 5032, https://doi.org/10.1038/s41598-024-54910-3.
[56] C. Cimini, A. Lagorio, S. Cavalieri, Development and application of a maturity model for Industrial Agile Working, Comput. Ind. Eng. 188 (2024) 109877, https://doi.org/10.1016/j.cie.2023.109877.